Compound microscopes are optical microscopes because the use of light rays for viewing the objects or specimens'.

- **In scanning electron microscope**, the whole specimen is scanned by a beam of electrons and an image is created by electrons reflected from the surface of the specimen.
  Scanning electron microscope shows the surfaces of structures, greater depths of focus(i.e a large part of the specimen is in focus at the same time) and three dimensional appreciation of the object.
- **A transmission electron microscope** is one in which the electron beam is transmitted through the specimen before viewing

∴ ***Answer C***

29. Which of the following mechanisms do cells use to regulate enzyme catalyzed reactions in metabolic pathways?

A .Enzyme denaturation
B .Irreversible inhibition.
C. End product inhibition
D. Competitive inhibition.

Explanation :

- **Enzymes are destroyed by heating:** This is because enzymes are made up of **proteins,** which can easily get destroyed by heat, High temperature (above 40°C)destroy the organization structure of the protein which makes the enzyme unable to carry out its normal functions. This process is known as **enzyme denaturation.**
- **Irreversible inhibition**: Irreversible inhibitors bind tightly and permanently to enzymes and destroy their catalytic properties. These drastic effects occur at very low concentration of inhibitors, and such substances are described as poisons.
- **End product inhibition**: Many metabolic reactions in the body are not single –step reactions ,but rather involve series of reactions. Such reactions are called **biochemical path** ways because they consists of a number of Sequentially interelated chemical reactions.
- **Competitive inhibition:** Competitive inhibitors have molecules with shapes that are complementary to all, or part of the active site of an enzyme.

∴ ***Answer C***

30. Which of the following modes of material transport across the cell membrane is **NOT** governed by the concentration gradient of the transported material?

A .Simple diffusion
B .Facilitated diffusion
C .Osmosis
D . Active transport.

Explanation :

- **Simple diffusion** is the movement of molecules or ions from a region of higher concentration to a region of lower concentration down the

diffusion gradient. The process is passive , that is , it does not require energy and happens spontaneously.

- **Facilitated diffusion** is essentially the same process as diffusion , in that it depends on a concentration gradient to allow particles to cross the membrane. However, it differs in that particles must be helped to diffuse across the membrane (their diffusion must be facilitated) by carrier protein or a channel protein within an ion pore.
- **Osmosis** is the process by which water moves across a partially permeable membrane. It is effectively the diffusion of water molecules from a region of their higher concentration to a region of lower concentration.
- **Active transport:** these require energy from the cells metabolism in the form of ATP to drive the molecules or ions from a region of lower concentration to a region of higher concentration.

∴ ***Answer D***

31. Which one of the following is **NOT** one of the roles that microorganisms play in ecosystems?

A .Nutrient recycling
B .Carbon fixation
C. Nitrogen fixation
D. Energy recycling

Explanation :

- **Nutrient recycling:** The elements needed to form the molecules of which organisms are made, remain in circulation when molecules pass from one trophic level to the next.
- **Carbon fixation (CO_2 fixation)** :In the dark phase of photosynthesis that carbon atom in the carbon dioxide is reduced to a glucose molecule using the energy generated in the light reaction. This is known as **carbon (carbon dioxide) fixation.**
- **Nitrogen fixation** is a part of nitrogen cycle in which the bacteria called nitrogen fixing bacteria found in the root nodules of legumes convert nitrogen gas in to ammonia which later on is transformed in to nitrate.
- Micro- organisms are not involved in energy recycling because there is no energy recycling in any ecosystem. Energy flows from the source, the sun to the plants and then to the consumers. Finally the energy is lost from the ecosystem as heat.

✓ **Carbon fixation** and **energy recycling** are not the roles played by micro-organisms.

∴ ***Answer B and D***

32. Which one of the following demographic factors affects the number of human population globally?

A . Natality
B . Migration
C .Emigration
D. Immigration

Explanation :

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- **Natality:** if natality exceeds mortality (more are born than die), the population numbers will increase.
- **Migration:** Movement in to (immigration) and out of (emigration)the area.
- **Emigration:** if emigration exceeds immigration (more leave than enter), the population number will decrease.
- **Immigration:** if immigration exceeds emigration (more enter than leave), the population numbers will increase.

∴ ***Answer B***

33. Which of the following is the best way to check whether an individual having a dominant phenotype is homozygous or heterozygous for the trait?
A .To self the individual.
B .To cross it to a heterozygous individual.
C .To cross it to homozygous recessive individual.
D .To cross it to homozygous dominant individual.

Explanation : To check whether an individual having a dominant phenotype is homozygous or heterozygous, the individual is crossed with another individual showing homozygous recessive characteristic. This technique of breeding is called **test cross or back cross.**

✓If the individual is homozygous for the dominant trait all of the off spring produce will show the dominant trait.

✓If the individual is heterozygous 50% of the off spring produced will show the dominant and the other 50% will show the recessive trait

∴ ***Answer C***

34. Which of the following ideas is **NOT** a part of Darwin's Theory of Evolution?
A .Over reproduction.
B .Use- and – disuse of body parts.
C .Existence of heritable variation.
D .Competition for scarce resources.

Explanation :

- **Over production :** all species tend to produce more off spring that can possibly survive. This is the part of Darwin's theory of evolution.
- **Use and disuse : Lamark** suggests that by continually using a structure or process, will be become enlarged or more developed. Conversely, any structure or process that is not used or is little used will become reduced in size or less developed.
- **Existence of heritable variations: Charles Darwin** after visiting the Galapagos island he concluded that there is variation among the off spring.

- From his observation **Charles Darwin** deduced that there will be "a struggle for existence" that is competition for scarce resources between members of the same species (because they over- reproduce, and resources are limited)

∴ ***Answer B***

35. Which of the following types of movements in response to a stimulus has no specific direction?
A .Taxis
B .Kinesis
C. Gravitropism
D. Phototropism

Explanation :

- **Taxis** in which the animal moves along a gradient of intensity of the stimulus towards the greatest intensity of the stimulus (**a positive taxis**) and sometimes away from the greatest intensity (**negative taxis**);there is a direction response to a directional stimulus
- **Kinesis** in which a change in the intensity of the stimulus brings about a change in the rate of movement, not a change in the direction of movement.
- **Gravitropism :** plant stems grow up wards and roots grow more or less down wards. The unidirectional stimulus producing this response is gravity. The response by plants to gravity is called **gravitytropism.**
- **Phototropism :** plant shoots grow towards the greater light intensity. This behavior is called phototropism.

∴ ***Answer B***

36. Which of the following laboratory equipment is used to separate the organelles of the cell according to their density?
A .Incubator
B .Centrifuge.
C. Measuring cylinder
D. Filter paper with fine pores.

Explanation :

- **Incubator** is an apparatus used to hatch eggs by means of heat artificially or growth of bacteria culture by keeping it for some time at an optimum temperature.
- **Centrifuges** are used to separate solids from liquids when simple filtration is not adequate for the task. Some centrifuges, called **ultracentrifuges** are used to separate the various components of animal and plant cells.
- **Measuring cylinder** is a graduate glass or plastic cylinder to measure volume of liquid.
- **Filter papers with fine pores** is one of the important laboratory materials used to filter suspended solid materials in water suspension

∴ ***Answer B***

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37. In the process of amino acid condensation, which one of the following happens?
A .Oxygen is used up
B .Carbon dioxide is released
C. Water is released as a byproduct.
D. Protein is broken down into amino acids.
Explanation : Two amino acids can be joined together by condensation to form a **dipeptide bond.**
✓ This takes place in much the same way as when two monosaccharide's join to form a disaccharide. Hand OH are lost to create a molecule of water. The H is lost from the amino group on one amino acid and the OH is lost from carboxyl group on the other amino acid.
∴ ***Answer C***

38. In which one of the following points does the induced- fit model of enzyme action differ from the lock- and - key model?
A .Enzymes lower the energy of activation.
B .Substrates bind at the active site of the enzymes.
C . During the reaction, an enzymes – substrate complex is formed.
D .The shapes of the substrate and active site are complementary.
Explanation :The **induced fit model suggests** that active site and the substrate are not naturally complementary in shape, but the binding of substrate molecules produces a conformational change (change in shape in the active site).This allows the substrate and active site to bind fully.
✓ The **lock – and –key model proposes** that the shapes of the substrate molecules are complementary to that of the active site rather like the shape of a key is complementary to that of the lock it fits.
∴ ***Answer D***

39. In an experiment that is testing the effect of temperature on the germination rate of certain seeds, in which one of the following should the experimental and the control group differ?
A .The amount of water they receive.
B .The age of the seeds assigned to them.
C .The number of seeds assigned to them.
D .The temperature at which they are kept.
Explanation :In an experiment that is investigating the effect of temperature on the germination rate of certain seeds, the same number of the seeds, are planted in same sized pots containing the same amount and the same type of soil. They are then supplied with the same amount of water at the same time. Since the investigation is aimed to find out the effect of temperature, the pots are placed in conditions of different temperature. Then the number of seeds germinated in each pot will be

counted after sometimes. The only difference between the pots containing seeds is temperature.
∴ ***Answer D***

40. Which of the following processes of photosynthesis does **NOT** require the presence of light to take place?
A .The splitting of water.
B .ATP formation
C .Reduction of NADP.
D. Carbon fixation.
Explanation : **Biologists** now know that there are two main sets of reaction in photosynthesis. They are known as the **light reactions** and the **dark reactions.**
✓ The **light reactions** involves a series of changes dependent on light. During these reactions, light energy is absorbed by chlorophyll and carotene molecules. Part of the absorbed energy is used to split water in to hydrogen and oxygen. Then by series of steps the other part of the absorbed energy is used to form ATP and reduce NADP molecules for later use during the dark reaction.
✓ **Dark reaction** of photosynthesis does not require light energy. It is the event that follows the light reaction where the carbon in the carbon dioxide is fixed or reduced to glucose using energy from light reaction (ATP and $NADPH_2$).
∴ ***Answer D***

41. In which part of the cell do Gram- positive and Gram- negative bacteria differ regarding their staining property with Gram's stain?
A .Cytoplasm
B .Cell membrane
C .Cell wall
D. Nucleus.
Explanation : Bacteria can be classified in to two main groups as **Gram positive and Gram negative.**
✓ **Gram-positive** – these bacteria are stained purple by Gram's stain.
✓ **Gram – negative** – these bacteria are stained pink by Gram's stain.
Because Gram's stain produces different results with different types of bacteria, it is called differential stain.
✓ The differentce is due to the structure of the cell wall of the different bacteria. Gram negative bacteria have much less peptidoglycan in their cell walls. This is the part of the wall that absorbs the stain .They have a membrane outside the peptidoglycan cell wall, which gram- positive bacteria do not have.
∴ ***Answer C***

42. When do populations of living organisms show exponential growth?
A .When the resources are plentiful.
B .Whenever they enter a new environment.
C .When they face strong competition from other species.
D .When the carrying capacity of the environment is reached.

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Explanation : In exponential growth, the population doubles in each time period, producing an ever increasing growth rate that is clearly not sustainable in nature. This occurs when all the organisms are adapted and reproduce rapidly due to plentiful resources.

∴ ***Answer A***

43. Which of the following terrestrial biomes experiences hot days and cold nights?

A .Tundra
B .Tropical rainforest
C. Desert
D. Grass lands

Explanation :

- **Tundra:** is an enormous biome, extremely uniform in appearance. The land usually remains frozen except the upper most 10 -20 cm permanent ice, usually exist within a meter of the surface.
- **Tropical rain forest** is occur near the equator warmth and moisture due to the abundant rain fall (2000 – 4500 mm per year)allow a continuous growth of plants.
- **Desert :** A biome where the average amount of precipitation is erratic and less than 25 cm per annum, and evaporation exceeds precipitation. Such areas have spares highly adapted vegetation, **Example:** Cacti, Succulent and spiny shrubs. Hot deserts such as Sahara have very high day temperature and very cold night.
- **Grass land :** This biome occupies quite large areas in both the temperate and tropical regions of the world. Grass lands provided good conditions for grazing animals. They are also very productive when converted for agricultural purpose because the soil is very rich in humus due to the rapid growth and decay of plants

∴ ***Answer C***

44.In DNA cloning technology, which of the following molecules serves as a vector of gene of interest to be transferred to bacteria host?

A .Bacteria DNA
B .Plasmid DNA
C. Nuclear DNA
D. Mitochondrial DNA.

Explanation :In DNA cloning technology, Genes can be transferred in to other cells using vectors. These are usually either plasmids (small pieces of circular DNA found in bacteria), or viruses

∴ ***Answer B***

45.To which of the following groups does HIV belong?

A .plasmids
B .Retroviruses
C.DNA viruses
D . Bacteriophages.

Explanation :

- **Plasmids** are small pieces of circular DNA found in bacteria. It is extra chromosomal DNA of the bacterial cell. That is different from the main genetic material of the cell.

- **Retro viruses** are viruses that contain RNA , but replicate in different way. When they infect cells, they release in to the cells their RNA and an enzyme that causes it to be **"reverse- transcribed"** in to DNA . Therefore ,HIV viruses are retro viruses.
- **DNA viruses** contain DNA as their genetic material.
- Bacteriophages are viruses that use bacterial cells as host cells to replicate their genetic material- DNA and synthesise viral proteins.

∴ ***Answer B***

46.Which of the following properties of water makes sweat an effective body cooler?

A .Its high specific heat
B .Its high surface tension
C. Its low density when frozen.
D. Its high heat of vaporization.

Explanation :

- **High specific heat**: the heat capacity of water is the amount of heat required to raise the temperature of 1 kg of water by 1°C.Water has a high specific heat
- **High surface tension**: The water molecules in the main body of mass of water are hydrogen bonded to other water molecules on all sides.
- **Low density when frozen :** is unusual for the solid form of a substance to float on the liquid form of the same substance, but ice floats on water.
- **High heat of vaporization**: This means that it takes a lot of energy to turn liquid water in to vapor (or steam).In turn this means that water does not vaporize too easily and that ponds do not dry too quickly in hot weather.

When we sweat, the energy needed to vaporize the sweat comes from our bodies. This is then lost from our bodies and so we cool down.

∴ ***Answer D***

47.Which of the following is **NOT** in agreement with the scientific method?

A .Putting forward testable hypotheses.
B .Carrying out experiments in duplicates.
C .Putting forward personal value judgments.
D .Analyzing results and drawing conclusions.

Explanation : The steps in the scientific method used to investigating a certain problem are as follows.

1 .**Observation** :the detail study or viewing of the natural word and accurate recording the sensory data and organization of these data in to concepts.

2 .**Formulation a hypothesis:** based on the observation made, the investigator forms a hypothesis or a suggested explanation of certain observed phenomena.

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3. **The formulated hypothesis** is tested by experimentation. Experiments are carried out to reject or accept the hypothesis.
4. **The experimental results** are analyzed and conclusion are drown.

∴ ***Answer C***

48. To which major area of relevance and applications of biology is the production of transgenic organisms related?

A .Agriculture
B .Environment
C. Biotechnology
D. Medicine

Explanation : A Transgenic organism is a genetically modified organism that contains a gene or genes transferred from another organism belonging to a different species. Transgenic organisms are produced by the modern bio technology known **as genetic engineering**

∴ ***Answer C***

49. What are the individual strands of a fungal mycelium called?

A .Fungus
B .Spores
C. Hyphae
D. Hydra

Explanation : The body of higher fungi is composed of threads or tube like structures called **hyphae**. An inter connected mass of these hyphae is termed the **mycelium.**

∴ ***Answer C***

50. What is the main mode of transmission of diseases such as cholera and typhoid fever?

A .Sexual intercourse
B .Bites of animal vectors
C. Blood – to - blood contact.
D. Drinking contaminated water.

Explanation:

- **Sexual inter course:** Organisms infecting the sex organs can be passed from one sexual partner to another during intercourse. Some of the examples are candidiasis, **syphilis, AIDS and gonorrhea.**
- **Bites of animal vectors:** many diseases are spread through the bites of insects. **Mosquitoes** spread malaria and testes flies spread sleeping sickness.
- **Blood- to – blood contact :** many of the sexually transmitted diseases can be transmitted by blood to –blood contact. Drug users sharing an infected needle can transmit AIDS.
- **Drinking contaminated water:** when unclean water containing the micro- organisms is drunk they colonize a suitable area of the gut and reproduce. They are passed out with faeces and find their way back in to the water. **Cholera and typhoid fever** are good examples of such diseases.

∴ ***Answer D***

51. Viruses are better characterized as:

A .Decomposers.
B .Producers.
C. Carnivores
D. Parasites.

Explanation: **Viruses** are sometimes referred to as micro-organisms ,although some biologists say that , strickly, they are not organisms at all. **Viruses** cannot independently carry out any of the processes common to all living organisms. They can only reproduce inside other cells. So they are all **parasites.**

∴ ***Answer D***

52. Which means of particle transport requires input of energy by the cell?

A .Simple diffusion
B . Facilitated diffusion
C .Osmosis
D. Active transport.

Explanation: Refer question 30 in 2004 year.

∴ ***Answer D***

53. Which of the following is NOT one of the stages in cellular respiration?

A .Calvin cycle
B . Glycolysis
C .Electron transport
D .Krebs cycle.

Explanation:

- **Calvin cycle (the three- carbon path way):** the reduction of carbon takes place in the stroma in a cycle called the calvin cycle. The Calvin cycle occurs in the dark reaction of photosynthesis.
- **Glycolysis:** is the anaerobic stage of aerobic respiration. Glycolysis is anaerobic break down of carbohydrate, specifically glucose to pyruvic acid in aerobic respiration.
- **Electron transport** is one of the stage of aerobic respiration that requires oxygen. The electron transport chain is a system of electron carries embedded in the inner membrane of the mitochondrion.
- **Krebs cycle** is one of the stage of cellular respiration that follow glycolysis. It is a cyclic reaction where the acetyl coenzyme a produced from pyruvate in a link reaction enters the cycle. In this cycle a number of series of chemical reactions occur that generate carbon dioxide, reduced NAD and FAD.

∴ ***Answer A***

54. Among the molecules found in cells, which of the following contains less energy?

A .A glucose molecule
B .An amino acid
C.A triglyceride
D. A water molecule.

Explanation:

- **Glucose molecule:** an oxidation of glucose in aerobic respiration produces 36 to 38 ATP molecules (the immediate source of energy for all living things)

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❖ **An amino acid** is an organic molecule that can generate a certain amount of ATP in reactions channeled to energy releasing path way.
❖ **Triglyceride** can also generate energy in the same way as amino acids, because it is an organic compound.
❖ **Water** is the result of (by product) energy releasing process. Its oxidation does not release energy.

∴ ***Answer D***

55. What is the correct equation for cellular respiration?
A .$6CO_2 + 6H_2O + Energy = 6O_2 + C_6H_{12}O_6$
B.$6O_2 + C_6H_{12}O_6 = 6CO_2 + 6H_2O$ +Energy.
C .$6O_2 + C_6H_{12}O_6 + Energy = 6CO_2 + 6H_2O$
D. $6CO_2 + 6H_2O = 6O_2 + C_6H_{12}O_6 + Energy$.
Explanation: The equation in choice A is summary of the process of **photosynthesis.**
✓ The equation in choice B is summary of cellular respiration known as **aerobic respiration.**
✓ The equations in choices C and D represent no known biochemical process in living organisms.

∴ ***Answer B***

56. What amount of net gain in ATP does glycolysis provided to a cell?
A .2 ATP molecules. C.18ATP molecules.
B .4 ATP molecules. D.36 ATP molecules.
Explanation: **Glycolysis** is the anaerobic stage of aerobic respiration which occurs in the cytoplasm. In this stage glucose is broken down to two pyruvate generating only 2 net ATP molecules.

∴ ***Answer A***

57. What is the term for the process by which organisms keep their internal conditions at a fairly constant state?
A .Catabolism C. Homeostasis
B .Evolution D. Photosynthesis.
Explanation:
❖ **Catabolism:** metabolic reaction involved in the breakdown of complex molecules to simple compounds.
❖ **Evolution:** in biology ,descent with modification, the process whereby all living things on earth have diverged, by direct descent from a single original life that occurred more than 3,000 million years ago.
❖ **Homeostasis:** the maintenance of constant internal environment by an organism.
❖ **Photosynthesis:** the synthesis of organic compounds from inorganic compounds (H_2O and CO_2) using light energy absorbed by chlorophyll.

∴ ***Answer C***

58. Which of the following requires expenditure of ATP?
A. Osmosis C. Simple diffusion
B .Facilitated diffusion D. Endocytosis
Explanation:
✓ Refer to question number 30 for the explanation of the terms, **Osmosis, Facilitated diffusion and simple diffusion.**
✓ **Endocytosis** is an active process involving the bulk transport of materials in to cells. It occurs by in folding or extension of cell surface membrane to form a vesicle or vacuole.
✓ ***Osmosis ,Facilitated diffusion*** and **simple diffusion** are passive transports processes that do not require the expenditure of ATP while **Endocytosis** is an active process that requires the expenditure of energy.

∴ ***Answer D***

59. What is the most probable selection pressure responsible for the evolution of green skin colour in frogs inhabiting tropical rainforests?
A .Climate C. Infection by pathogens
B .Reproduction D. Predation.
Explanation: The most probable selection pressure responsible for the evolution of green- skin colour in frogs inhabited in tropical rain forest is predation.

∴ ***Answer D***

60. Which group of organisms found in Ethiopia is represented by the highest number of endemic taxa ?
A .Mammals C. Birds
B .Amphibians D. Plants
Explanation:
❖ **Mammals :** 22 species are endemic to Ethiopia.
❖ **Amphibians :**17 species of amphibians are endemic to Ethiopia.
❖ **Birds :** 27 species of birds are endemic to Ethiopia.
❖ **Plants :** approximately 1150 plant species are endemic to Ethiopia

∴ ***Answer D***

61. Why are fossils of soft bodied organisms usually relatively rare in the environment?
A .They are generally small in size.
B .Their bodies decompose readily.
C .They all lived in environments where sedimentation did not occur.
D .They were never common in environments in which they lived.
Explanation: Clearly death of the organism, for some reason or other is the first stage of fossil formation. The animal or a plant must die in or so close to the water that is covered by water immediately, after , or shortly after, death. The water insulates the remains from many of the elements

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that contribute to decomposition. Bacteria will still decay the soft body parts over a long period but leave any hard parts unaltered. This makes very difficult for soft bodies organisms to be fossilized .That is why the fossils of soft bodied animals are relatively rare.

∴ ***Answer B***

62. Which of the following is true about the evolutionary origin of groups of organisms?
A . The dinosaurs appeared before the origin of the land plants.
B .The earliest homo sapiens appeared before the flowering plants.
C .The first photosynthetic organism appeared before the oldest eukaryotes.
D .The first animals appeared before the formation of free O_2 in the atmosphere.

Explanation: According to the evidence from the evolutionary tree of animals and plants.

❖ The first vascular land plant appeared between 400-450 million years ago.
✓ The **dinosaurs** appeared between 200 to 250 million years ago.
❖ The first flowering plants appeared between 50 to 150 million years ago.
✓ The earliest **homosapines** appeared about 1 million years ago.
❖ The first **photosynthetic organism** (cyno bacteria) appeared some 3.5 million years ago.
✓ **Eukaryotes** appeared between 1.9 to 2.1 million years ago.
❖ The process of photosynthesis evolved 3.5 million years ago. But the concentration of oxygen began to rise 2.4 billion years ago. The oxygen produced by the photo-autotrophs had made it possible for aerobic respiration to evolve an energy – releasing pathway. As this process releases for more energy than does the anaerobic path way more active organisms could now evolve- the animals perhaps 600 to 700 million years ago.

∴ ***Answer C***

63. A cell was examined under a compound light microscope with an eyepiece lens marked 13 X and an objective lens marked 40X .How many times larger would the cell appear to the viewer?
A .40 times B .53 times C .400 times D .520 times.

Explanation: Magnification of a microscope = the magnification power of an eye piece (ocular) lens × the magnification of an objective lens.
The magnification of this specific cell = eye piece lens 13 X × objective lens 40 X = 520 X

∴ ***Answer D***

64. How many moles of ATP will be generated as a result of the oxidation of one mole of $FADH_2$ in actively respiring mitochondrion?
A. 0 B.3 C .2 D .6

Explanation: **FAD (Flavine Adenine dinucleotide**)is a coenzyme that transfers hydrogen and electron from the Krebs cycle(a stage of aerobic respiration) to the electron transport chain (the last stage of aerobic respiration).
✓ It is known as a **hydrogen carrier.**
✓ The transfer of hydrogen and electrons in the electron transport chain in the mitochondrion generates ATP. The transfer of hydrogen's and electrons from a single $FADH_2$ (the reduced form of FAD) generates 2 ATP molecules.

∴ ***Answer C***

65. Which alternatives contains only crops known to have been domesticated within Ethiopia?
A .Guizotia Abyssinica, Zea Mays ,pisum sativum.
B .Vicia faba, carica papaya,Musa paradisiaca.
C .Oriza sativa ,Triticum aestivum, solanum tuberosum.
D .Coffea Arabica, Eragrostis tef, Ensete ventricosulum.

Explanation: Based on evidence from archaeological finding and ancient literature , it is believed that crop plants have initially originated and evolved from the wild species. It is believed that indigenous crops such as Eragrostis tef, noug(Guizotia abyssinica), Ensete (Ensete ventricosulm), Coffe (coffee Arabica) , archote (cocci nian Abyssinia), Oromo / Welayta dinich(plectranthus edulis) were domensticated in Ethiopia.

∴ ***Answer D***

66. Which of the following statements is true about the nitrogen cycle?
A .Plants fix nitrates from atmospheric nitrogen.
B .The nitrogen used by animals largely comes from plants.
C .Nitrogen is consumed by bacteria and removed from the soil.
D .Nitrogen- fixing bacteria reduce the total amount of available nitrogen.

Explanation: In the nitrogen cycle-decomposers (bacteria and fungi)decomposes the dead organic matter from plants, animals and execratory products which contain nitrogen in to ammonium ions.
✓**Nitrifying bacteria** oxides ammonium ions to nitrates (which are taken by plants)in the process called nitrification.
✓**Denitrifying bacteria** reduce nitrate to nitrogen gas that escapes from the soil. This reduces the total amount o f nitrogen available to the plants and therefore , to all the other organisms also.

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✓**Nitrogen – Fixing bacteria** "fix" nitrogen gas in to ammonium ions which is then converted to nitrates.
✓**Nitrates** are taken from the soil by plants to make nitrogen containing compounds such as **proteins**.
✓The plants are eaten by animals and as they do so they pass the nitrogen containing compounds to the **animals**.

∴ ***Answer B***

67. A genetic cross between two F1 – hybrid pea plants having yellow seeds(dominant)will yield what percent green-seeded (recessive)plants in the F2 generation?
A .0% B.25% C.50% D. 75%
Explanation: **A genetic cross** between two F_1- hybrid pea plants having yellow seed (dominant)will yield off spring in the F_2 generation in the ratio of 3:1 .That is 3 yellow seeded plants (the dominant) to 1 green seeded plants (recessive). On the other hand.
✓ ¾ of the plants will be yellow seeded (75%).
✓ ¼ of the plants will be green seeded the recessive(25%).

∴ ***Answer B***

68.Which of the following is true for cellular respiration?
A .Restricted to plant cells.
B .Restricted to animals cells.
C .Occurs in all eukaryotic cells
D. Occurs in prokaryotic cells only
Explanation: The cells of plants , animals , unicellular protozoan , fungi and algae are all **eukaryotic** .
All eukaryotic cells generate energy using cellular respiration or aerobic respiration. Aerobic respiration is more efficient energy releasing process than anaerobic respiration which occurs in prokaryotic.

∴ ***Answer C***

69.What could be the main reason behind the currently observed slow or stable rate of population growth in the industrialized countries?
A . Good family planning
B . Increasing death rate.
C .Poor health conditions.
D .High rate of child death.
Explanation: Most industrialized countries are in the final stage of demographic transition where the birth and death rates are low. This is the result of family planning good , health care improving status of women and later marriage. The overall result is a stable rate of population growth.

∴ ***Answer A***

70.What is the average projected rate of loss of biodiversity every 50 years?
A .5% B .10% C .20 % D .50%
Explanation: Our biodiversity has fallen from 90% of its potential in 1900, to 70% in 2000 and it is projected to fall to 60% by 2050 and carrying on the projection, to 50% by 2100,with most of this biodiversity in areas such as desert and tundra that are difficult to exploit. We are losing 10% of our biodiversity every 50 years.

∴ ***Answer B***

71.In an experiment designed to test the effect of different concentration of a fertilizer on the growth rate of a plant, which one of the following is the dependent variable?
A .The growth rate of the plant.
B .Plants assigned to the control group.
C .Concentrations of the fertilizer applied.
D .Plants assigned to the experimental group.
Explanation: The factor that the scientist changes or manipulate is called the **independent variable.**
✓The factor that the scientist measures to see if it changes when the independent variable is changing is called **the dependent variable.**
✓In this investigation designed to test the effect of different concentration of a fertilizer on the growth of the plants the growth rate of the plant is dependent on the concentration (amount) of fertilizer added. An increase or decrease in the concentration of the fertilizer applied determines the growth rate of the plant. Therefore, the growth rate of the plant that is measured by the investigator is the **dependent variable** and the amount of the fertilizer applied that is changed or manipulated by the is the investigator is the independent variable.

∴ ***Answer A***

72.In which one of their structural parts do different molecules of amino acids differ from one another?
A .In their R group. C. In their carboxyl group.
B .In their amino group. D. In their alpha- carbon group.
Explanation: **Amino acids** contain the COOH (carboxyl) group and an amino group (NH_2). Both the COOH and the NH_2 group are attached to the same carbon atom designated as " the alpha carbon".
✓The various amino acids differs in their side **chains, or "R " groups.** " R" may be simple, as in alanine where it is only methyl group (CH_3), or it may be very complex , as in tryptophan, where it includes two ring structures.

∴ ***Answer A***

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73. Suppose 25 % of the molecules of an enzyme are inhibited by a non-competitive inhibitor,which one of the following would happen if the amount of the substrate is increased by 50%?

A .The reaction rate would double.
B .More enzymes molecules would get inhibited.
C .The rate of the reaction would decrease by 50%.
D .The rate of the reaction would remain unchanged.

Explanation: **Non – Competitive inhibitors** do not compete for the active site. Instead , they bind to another part of the enzyme called **allosteric site.** This produces conformational change in the part of the enzyme molecule that includes the active site. Because of this the active site is a different shape and can no longer bind with the substrate to catalyze the reaction. The effectiveness of a non- competitive inhibitor is in no way affected by the concentration of the substrate. In this particular case where 25% of the enzyme molecules are inhibited by non-competitive inhibitor an increase in the concentration of the substrate by 50% would not change the rate of the reaction , because the effectiveness of the non-competitive inhibitor is not affected by substrate concentration

∴ ***Answer D***

74. Suppose we consider four hypothetical cells(designated A,B,C,and D) having cubic shape with their sides measuring 2,4,6 and 8 arbitrary units, respectively which of these cells has the largest surface area to volume ratio?

A .Cell A B .Cell B C. Cell C D. Cell D

Explanation: **Cell A- arbitrary unit – 2**

✓ The area of one side of cell A is
$= 2\ a.u \times 2\ a.u$
$= 4a\ .u^2$

✓ There are 6 sides of the cell
= The surface area of cell A is
$= 6\times 4a.u^2$
$= 24a.\ u^2$

✓ The volume of cell A is length × breadth × height.
$= 2\ a.\ u \times 2\ a.u \times 2\ a.u$
$= 8a.\ u^3$

✓ The surface area to volume ratio of cell A
$= \frac{24}{8} = 3 = 3:1$

Cell B- arbitrary unit 4.

✓ Area of one side = 4 a.u ×4 a.u
$= 16a.\ u^2$

✓ The surface area = 16 a . u^2 × 6
$= 96a\ .\ u^2$

✓ The volume = 4 a.u× 4 a.u× 4a..u
$= 64\ a.U^3$

✓ The surface area to volume ratio of cell B 1.5:1

Cell C- arbitrary unit 6

✓ The area of one side = 6 a.u × 6 a .u
$= 36\ a.\ u^2$

The surface area = 36 a .u^2 × 6
$= 216\ a\ .u^2$

The volume = 6 a.u× 6a.u ×6 a.u
$= 216\ a.\ u^3$

The surface area to volume ratio is 1: 1

Cell D- arbitrary unit – 8

The area of one side = 8 a.u ×8 a.u
$= 64\ a.u^2$

The surface area = 64 $a.u^2$× 6
The surface area = 384 a .U^2
The volume = 8 a.u× 8 a.u ×8 a.u
$= 384\ a\ .u^3$

The surface area to volume ratio of cell D is 1: 1

∴ ***Answer A***

75. In cyclic photophosphorilation, what is the source of the recycled electron?

A .Reduced NADP.
B. Chlorophyll molecule.
C. Adenosine triphosphate.
D. Photolysis of water molecules.

Explanation: Plants sometimes generate ATP by cyclic photophosphorilation. In cyclic photophosphorilation, only photo system I is used. No oxygen and no reduced NADP are formed , because there is no photolysis of water (splitting water into hydrogen and oxygen)which usually occurs in photo system II . Electrons lost from the chlorophyll molecules are returned to it. Hence the name cyclic. It is the chlorophyll that generates the electrons required to synthesis ATP. This process usually only happens when sugars cannot be synthesized for some reason- such as lack of carbon dioxide.

∴ ***Answer B***

76. Which one of the following steps in the life cycle of HIV is blocked if an antiretroviral drug that inhibits the reverse transcription enzymes is given to an AIDS patient?

A .Formation of DNA from RNA.

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B .The entry of HIV into CD4 cells.
C .The assembly of parts into HIV particle.
D .The integration of HIV DNA into host chromosome.

Explanation:

- The antiretroviral drug that inhibits the action of reverse transcriptase enzyme inhibits the conversion of RNA in to DNA. Because reverse transcriptase is the enzyme that converts RNA in to DNA .This is one of the vital process in the reproduction of HIV - particle.
- The HIV – particle enters the T- helper cell or CD4 cell when the gp120 on the surface of HIV binds with the CD4 – protein receptor on the surface of T – helper cell. Therefore, ant retro viral drugs that inhibits the binding between gp 120 and CD_4 protein receptor inhibits the entery of HIV in to CD_4 .cell.
- **Protease** cuts protein chains in to protein parts for HIV particle, which can assemble and "bud" from the T- helper cells. Antiretroviral drugs that inhibits the action of protease inhibits the assembly of the parts in to HIV particle.
- The antiretroviral drug that inhibits the action of enzyme that incorporate the HIV DNA within the host cell nucleus into the host cell chromosomes inhibits the integration HIV DNA in to the host cell DNA.

∴ ***Answer A***

77. Carbon 14 has a half life of about 5730 years. Suppose a fossil contains only 12% of the amount of carbon 14 normally present in living organisms, how old is the fossil?
A .5730 years.
B .11460 years.
C. 17190 years
D. 22920 years.

Explanation: Carbon 14 has a half life of about 5730 years. So after 5730 years only 50% of the original carbon 14 atoms will remain.In the next half life only 25% of the original carbon 14 remains. That is 2 half lives about 11460 years.
In the third half life only 12.5 % of the original carbon 14 remains, that is about 17190 years.
There for in a fossil where only 12% of the original carbon atoms remained , the fossil is 17190 years old.

∴ ***Answer C***

78. What is the reason that Louis Pasteur used a swan- necked flask instead of straight necked one in his experiment designed to disprove the theory of spontaneous generation?
A .To allow free passage of air to the broth inside the flask.
B .To prevent the escape of any microorganism from the flask.
C .To keep the broth in the flask hot to kill microorganisms.

D .To trap particles from the air that might enter the flask before reaching the broth.

Explanation: **Louis pasteur** was a scientist who showed that micro-organisms can only arise from pre-existing organisms disproving the theory of spontaneous generation. In his experiment he used a swan-necked flask (**an S- shaped neck**),So that the bacteria and dust particles settle or trapped in the bend of the swan necked flask. This prevented the passage of micro- organisms and dust particle down in to the broth (a sterilized medium for the growth of bacteria).

∴ ***Answer D***

79. If one mixes a sample of a fruit juice and some drops of Benedict's solution and obtains a brick- red precipitate upon warming the mixture, what does the juice contain?
A .Starch
B .Reducing sugar
C. Sucrose
D. Protein

Explanation:

- The **presence of starch** in any food stuff is tested using iodine solution (potassium iodide).If the food stuff contains starch it turns blue- black. This known as iodine test for starch.
- **The presence of reducing sugar**, such as glucose in a food stuff including fruit juice can be tested using the benedict's test for reducing sugars. When a mixture of the food stuff and blue- Benedict's solution is heated and if a brick- precipitate is formed , then it proves that the food stuff or juice contained reducing sugar.
- **Sucrose** is a non-reducing sugar and its presence in a food stuff is tested using benedict's test for non-reducing sugars.
- **Protein** is tested using a biuret test. When a mixture of a food stuff, sodium hydroxide solution and cooper sulphate is allowed to stand for few minutes, and if the mixture produces purple colour it proves that the food stuff contained protein.

∴ ***Answer B***

80. In competitive inhibition, which one of the following factors determines the rate of the inhibition?
A .The reaction temperature.
B .The enzyme concentration.
C .The substrate concentration.
D .The ratio of inhibitor to enzyme concentration.

Explanation: **Competitive inhibitors** have molecules with shapes that are complementary to all, or part , of the active site of the enzyme. They are often similar in shape to the substrate molecules. They can bind to the active site and prevent substrate molecules from binding. The binding is only temporary and the competitive inhibitor is quickly

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released. A competitive inhibitor blocks the active site so substrate molecules cannot bind. The overall effect on the rate of reaction depends on the relative concentration substrate and inhibitor molecules. An increase in the concentration of the substrate increases the chance of collision between the substrate and the enzyme. Hence increasing the chance of binding of the substrate to the active site of the enzyme.

∴ ***Answer C***

81. What type of molecules **CANNOT** pass across the cell membrane by simple diffusion?
A .Charged molecules.
B .Non- polar molecules.
C .Lipids soluble molecules.
D. Molecules of very small size.
Explanation: To pass through the plasma membrane by simple diffusion particles must be:
- Molecules of very small size
- Lipid soluble molecules
- Non-charged(non- polar) molecules.

❖ Charged particles cannot pass through the cell membrane by simple diffusion.

∴ ***Answer A***

82. Which of the following is true about the antiretroviral drugs currently used to treat AIDS patients?
A .They cure AIDS
B. They stop HIV transmission
C .They serve as anti-HIV vaccines.
D .They slow down HIV multiplication.
Explanation: Although there is no cure for AIDS and , as yet , no vaccine to give immunity against infection, there are a number of drugs called **anti-retroviral drugs**- that can be effective in slowing down the progression to AIDS .These drugs work by blocking the reproduction of the virus in the CD_4 lymphocytes.

∴ ***Answer D***

83. Among the following couples whose ABO blood genotypes are shown, which one can produce children of A ,B, AB and O blood types?
A .OO and AB
B.BO and AA.
C. BO and AO
D.BB and AO
Explanation:

❖ Parental blood type (phenoe type) –O and AB
Parental genotype I^OI^O and I^AI^B

Gametes I^O and I^A, I^B
Fertilization

	I^A	I^B
I^O	I^AI^O	I^BI^O
I^O	I^AI^O	I^BI^O

50% of the children have A blood type.
50% the the other children have B blood type.

❖ parental blood type (phenotype) B and A
Parental genotype I^BI^O and I^AI^A.
Gametes I^B, I^O and I^A
Fertilization

	I^B	I^O
I^A	I^AI^B	I^AI^O
I^A	I^AI^B	I^AI^O

50% of the children have **AB blood group**
50% the other children **A blood group.**

❖ Parental blood type (phenotype)B and A.
parental genotype I^BI^O and I^AI^O.
Gametes I^B, I^O and I^A , I^O.
Fertilization

	I^A	I^O
I^B	I^AI^B	I^BI^O
I^O	I^AI^O	I^OI^O

Children of A,B AB and O blood type are produced.

❖ Parental blood type B and A.
Parental genotype I^BI^B and I^AI^O.
Gametes I^B and I^A , I^O.
Fertilization

	I^A	I^O
I^B	I^AI^B	I^BI^O

Children of AB and B blood type are produced.

∴ ***Answer C***

84. Which of the following pairs of molecules can give information about how much two species are evolutionary related to one another
A .DNA and proteins.
B .Starch and cellulose.
C .Lipids and carbohydrates.
D. Carbohydrates and Proteins.
Explanation: .Chemicals that have been used in such analysis include:

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✓ **DNA** the base sequences of DNA from different organisms is compared.

✓ **Protein** such as cytochrome (found in the electron transport chain of respiration)and haemoglobin. The different in the amino acid sequence of cytochrome and the difference in the amino acid sequence of the four chains of hemoglobin are used to find out the evolutionary relationships of various organism.

∴ ***Answer A***

85. Suppose three potato cylinders are kept for some time in 15%, 8% and 4 %sucrose solutions, respectively, and the fourth cylinder is kept in distilled water, which of the cylibnders will be more flaccid?

A .The cylinder in 4% solution.
B .The cylinder in 8% solution.
C .The cylinder in 15% solution.
D .The cylinder in distilled water.

Explanation: If you put plant cells in water (weak solution)water moves by osmosis is and the **cell swells up** .The swelling protoplasm exert pressure called **turgor pressure** against the cell wall which eventually stops movement of water into the cell. When this point is reached we say the cell is turgid.

If you put a plant cell in to a solution which is stronger (concentrated solution) , water moves out of the cell and the cell **shrinks**. It loses its turgidity and become **flaccid**. This process is known as **plasmolysis**.

The greater the concentration of the solution is the more the cells flaccid. In this case the potato cylinder kept in 15% sucrose solution flaccid more , because 15% sucrose solution is the most concentrated solutions among the others.

∴ ***Answer C***

86.If there were no free oxygen to breathe, which one of the following steps of the respiration process can operate in our body?

A .Glycolysis
B .Krebs cycle
C .Electro transport chain
D. Reaction that links glycolysis &Krebs cycle

Explanation:

- **Glycolysis** is the anaerobic stage of aerobic respiration. It does not require oxygen and occurs in the cytoplasm.
- **Krebs cycle** is a cyclic series of reaction that occurs in the matrix of the mitochondrion. It is one of the stage of aerobic respiration that does not require oxygen. In this stage of respiration CO_2 ,reduced NAD and reduced FAD are generated.

- **Electron transport chain** involves a transfer of electrons from one electron carrier to another electron carrier. The electron is finally accepted by oxygen. Oxygen serves as final acceptor of electron. Electron transport chain is the last stage of aerobic respiration that generates most of the ATP molecules.
- **The reaction that links glycolysis and Krebs cycle known as link reaction does not require oxygen. It is a reaction in which pyruvate is converted to acetyl co- A.**

∴ ***Answer A***

87. Which stage in a primary ecological succession contains more biodiversity?

A .The third seral stage.
B .The second seral stage.
C .The climax community.
D. The pioneer community.

Explanation: **Succession** is a progressive series of changes in vegetation and animal life of an area from initial colonization to the final stabilized stage or, the climate.

- **The third seral stage** of primary succession contains more biodiversity than the second seral stage because organisms of the second seral stages changes a biotic conditions and this allows more species to colonize the area.
- **The second serial stage** contains more biodiversity than the first seral stage (but less than the third stage)because the pioneer organisms change the a biotic condition allowing the colonization of the area by more species.
- **The climax community** is the last stage of the primary succession. It is the stable community and the most complex state of the primary succession with maximum biodiversity.
- **The pioneer community** refers to the first group of organisms that first colonizer the uncolonized area .It has the least biodiversity.

∴ ***Answer C***

88.In the classical conditioning experiment performed by Pavlov on dogs, which of the following alternatives is the unconditioned stimulus?

A .The sound of the bell.
B .The smell of the food.
C .The salivation at the sound of the bell.
D. The salivation at the smell of the food.

Explanation:

- **The sound of the bell** is the conditional stimulus .The conditioned stimulus is the neutral stimulus that does not initially produce the unconditioned response. But after association with unconditioned stimulus , it triggers the same response.

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✓ The sound of the bell initially was not producing the unconditional response such as secretion of saliva alone.

❖ **The smell of the food** is unconditional stimulus. The unconditional stimulus is a stimulus that "**unconditionally** "naturally and automatically triggers a response. For example, when you smell food, you immediately feel very hungry (secrete saliva).

❖ **The salivation at the sound of the bell** is the conditioned response. It is a response to the previously neutral stimulus (the sound of the bell).

❖ **The salivation at the smell of the food** is unconditional response. It is the unlearned response that occurs naturally to the unconditioned stimulus(the smell of food). In the secretion of saliva example (feeling hunger) , Salivation (feeling hunger) is the unconditioned response.

∴ ***Answer B***

89. A biologist applied the scientific method repeatedly, gathered a large amount of supporting experimental data and finally described a pattern or relationship between different factors. What is the best term that refers to facts established in this way?

A .Theory
B .Hypothesis
C. Prediction
D. Law

Explanation:

❖ **Theory :** Hypothesis are often tested by experimentation. Any hypothesis that cannot be rejected is provisionally accepted .A hypothesis that has with stood many such tests (experimentations) and has been shown to allow prediction to be made is known as **theory**.

❖ **Hypothesis :** Observation provide the raw material, which leads to the formation of hypothesis. Hypothesis could be defined as a suggested explanation of certain observed phenomenon.

❖ **Prediction** is an educated guess as to how the biologist think his/her experiment will turn out.

❖ **Law** is a theory that may generate confidence through its predictive ability.

∴ ***Answer D***

90. What would most likely result if mitosis fails to be accompanied by cytoplasmic division?

A .Two cells without nuclei.
B .One cell without a nucleus.
C .Two cells each with one nucleus.
D. One cell with two identical nuclei.

Explanation: **Mitosis** is a nuclear division where two daughter nuclei produced are identical to one another. Usually nuclear division or mitosis is followed by division of the cytoplasm known as **cytokinesis** to form two identical daughter cells. If mitosis or nuclear divisor is not

followed by cytokines is (cytoplasmic division) or cytokinesis fails , a cell with two nuclei is produced.

∴ ***Answer D***

91. The tropical rainforest largely found in South America and Africa can be best characterized by a combination of which environmental features?

A .Low rainfall and low temperature
B .High rainfall and high temperature.
C .Low rainfall and high temperature.
D .High rainfall and low temperature.

Explanation: The **tropical rain forests** are largely found around the equator which is the main tropical region. It is a hot and wet region of the world. Therefore, **tropical rain forests** are characterized by a combination of high rain fall and high temperature.

∴ ***Answer B***

92. Which of the following terms mean stages in an ecological succession?

A .Pioneers
B .Climaxes
C .Seres
D. Niches

Explanation:

❖ **Pioneer** refers to the first group of organisms that first colonized the uncolonized area.

❖ **Climaxes** are the final or stable community in a succession of natural plant communities in one area under a particular set of conditions.

❖ **Seres :** The various stages in a succession are called **seres**.

❖ **Niches** : are the functional role of an organism is a community. If two species occupy the same niches then competition occurs until one has replaced by the other.

∴ ***Answer C***

93. What is the source of the oxygen that is produced during the process of photosynthesis by higher plants?

A .CO_2
B. H_2O
C. ATP
D. Chlorophyll

Explanation:

❖ In organic **CO_2** is a raw material in the process of photosynthesis. CO_2 is reduced to glucose using energy obtained in the light phase of photosynthesis.

❖ **H_2O** is another raw material in the process of photosynthesis. water is a source of hydrogen for synthesis of glucose and it is also a source of oxygen that is given off during the process of photosynthesis.

❖ **ATP**(Adenosine triphosphate) is the source of energy used to combine hydrogen's and carbon dioxide to make glucose.

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❖ **Chlorophyll** is a green pigment found in the cells of photosynthetic organisms. It absorbs light energy of the sun. It is structure where radiant energy is transformed in to chemical energy.

∴ ***Answer*** B

94. What does the structural similarity between the flippers of whales and arms of human show?

A .Whales evolved from the human species.
B .Whales are older than the human species.
C .The human species began life in the oceans.
D .Whales and humans had a common ancestry.

Explanation: Some organisms have anatomic structures that are very similar in form, but very different in function. We call such structures **Homologous structures**. Because they are so similar, they indicate an evolutionary relationship and a common ancestor of the species that posses them .perhaps the best known example of homologous structures is the fore **limb of mammals**. When examined closely , the fore limbs of humans, whales ,cats and bats are all very similar in structure. Each possesses the same number of bones arranged in almost the same way , while they have different features and function in different ways:

- arms for manipulation in humans.
- leg for running in cats.
- flipper for swimming in whales.
- wing for flying in bats.

∴ ***Answer D***

95. Which factors are involved in the determination of climax vegetation?

A .Temperature and precipitation.
B .Grazing and browsing animals.
C .Radiation and reflection.
D .Predators and preys.

Explanation: The main important factors that could determine the type of climax community formed include:

- temperature
- precipitation
- soil type
- soil depth

∴ ***Answer A***

96. Which parts of the angiosperm flower are both essential for the success of hybridization experiments?

A .Sepal and petal
B .Stamen and petal
C. Pollen and filament
D. Gynoecium and andoecium

Explanation:

❖ **Sepals** are often green and **petals** brightly coloured. These two specialized sterile parts of a flowers are said to be non-essential parts.
❖ **Stamens and carpels (pistils)** are the most essential and fertile parts which are involved in the formation of ovules which later become seeds.
✓ The stamens, the male reproductive organs collectively known as **androecium** and the carpel's (pistils), the female reproductive organs are collectively known as the **gynoecium**. All the explained structures in the above statements are parts of the angisperm (flowering plants) flower parts.

∴ ***Answer D***

97. Which of the following can be given as a good reason for finding large numbers of plant and mammal species in Ethiopia today?

A . Lack of ecological disturbance.
B . Environment free from predators.
C . Presence of many biomes and habitats.
D . Good ecological and biodiversity management.

Explanation: Ethiopia is an important regional centre of biological diversity. The wide range in **altitude and climate , the isolation** of the **high lands** of Ethiopia, and the fact that there are so many different biomes and habitats present in the country are some of the reasons for its high biodiversity

∴ ***Answer C***

98. How do human beings increase biodiversity?

A .By reducing species richness.
B .By increasing genetic variability.
C .By promoting habitat uniformity.
D .By narrowing ecological variability.

Explanation :

❖ **Species richness** is the number of different species in a particular area. Any factor affecting this species richness affects the biodiversity of that area or it reduces the biodiversity of the area.
❖ **Increasing genetic variability** is increasing the variability of the species of plants and animals. This increases the biodiversity of an area. Humans can increase genetic variability using various biotechnology techniques such as breeding and genetic engineering.
❖ **Promoting habitat uniformity** decrease the number of niches in the whole area .One of the measure of biodiversity is the ecological diversity of the species. The fewer the number of niches the less the biodiversity of the species.

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❖ **Narrowing ecological variability** means reducing the number of niches. This results a reduction in the variety of the species occupying the niches.

∴ ***Answer B***

99. What are the environmental advantage of using enzymes in industry?

A .It makes high production possible with less input of heat.

B .It makes high production possible with high input of heat.

C .It makes high production possible with emission of more CO_2.

D .It helps high production with supply of more heat and emission of more CO_2.

Explanation: One of the appeals of using enzymes in industry is that they allow the reactions involves in the processes to be carried out at much lower temperatures. This means less energy (and therefore less money) is spent on heating the reactions. Because less heating is required, less carbon dioxide is produced and this can benefit the environment as carbon dioxide is a green house gas and its accumulation in the atmosphere can lead to global warming.

∴ ***Answer A***

100. Why is it th... the woodlice are typically found under logs, stones, bark and amongst leaf litter?

A .To be sheltered in a dry windy environment.

B .To run away from the area where the air is humid.

C .To make sure that they are in the hottest place all the time.

D .To reduce the rate at which water is lost from their bodies.

Explanation: Wood lice are small land- dwelling crustaceans. There are many different species , but all are quite similar. Because of their flattened shape and small size ,they have a relatively large surface – area – to – volume ratio. This means that they lose water quickly through their body surface. This happens quickly because they have no waxy cuticle covering their bodies to limit loss of water .They are typically found under logs, stones , bark and amongst leaf litter. These areas all have a more humid atmosphere , which reduces the rate of water loss from the woodlice. They are also dark areas.

∴ ***Answer D***

Booklet code:31

1 . Which of the following steps in the scientific method comes following observation?

A .Prediction
B .Hypothesis
C. Experiment
D. Theory

Explanation:

❖ **Prediction** is an educated guess as to how the biologist thinks his/her experiment will turn out.
❖ **Hypothesis:** observations provide the raw material ,which leads to the formation of hypothesis. Hypothesis could be defined as suggested explanation of certain observed phenomena.
❖ **Experiment** :Hypothesis are often tested by experimentation.
❖ **Theory:** A hypothesis that has with stood many tests and has been shown to allow prediction to be made is known as **a theory.**

∴ ***Answer B***

2. The main component of the plant cell wall is _____ ?

A .Starch
B .Cellulose
C. Protein
D. Chitin

Explanation:

❖ **Starch :** is a polymer of ∝ - glucose.It is a major fuel store in plants , but is absent from animals where the equivalent is glycogen.
❖ **Cellulose** is a polymer of β – glucose unlike starch and glycogen it has a structural role. About 50% the carbon found in plants is in cellulose.
❖ **Protein :** All proteins contain four essential elements; **carbon , hydrogen , oxygen and nitrogen.** These elements are bonded together to form compounds called amino acids. Amino acids join to form protein molecules.
❖ **Chitin:** is closely related to cellulose in structure and function being a structural polysaccharide. It occurs in some fungi where its fibrous nature contributes to cell wall structure.

∴ ***Answer B***

3. Which of the following classes of enzymes digests carbohydrates?

A .Amylases
B .Lipases
C. Proteases
D. Nucleases

Explanation:

❖ **Amylase** is an enzyme that acts on starch (carbohydrate)to convert it to maltose (disaccharides).
❖ **Lipase** is an enzyme that changes fats and oils into fatty acids and glycerol's.
❖ **Proteases** is acts on protein to convert to peptides.

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❖ **Nucleases** is any of a class of enzymes that degrade nucleic acid (DNA or RNA) into shorter oligonucleotides or single nucleotide sub units by hydrolyzing sugar – phosphate bonds in the nucleic acid back bone.

∴ ***Answer A***

4. Which of the following kingdoms of life is consisting of prokaryotic organisms?
   A .Fungi
   B . Monera
   C. Protista
   D. Plantae.

Explanation:

❖ **Fungi** are eukaryotes organisms where their body is composed of threads or tube like structures called **hyphae.**

❖ **Monera** is includes organisms , the cells of which are prokaryotic.That is they lack a nuclear memebrane around their nuclear material.

❖ **Protista** is according to some biologists kingdom protista contains only unicellular eukaryotes. But other biologists include also multicellular algae with in this group.

❖ **Kingdom plantae :** they are eukaryotic multicelluar. They have a cell wall made up of cellulose and other polysaccharides.

∴ ***Answer B***

5. Which component of soil fertility is improved when farmers grow legumes in crop rotation?
   A .Phosphorus
   B .Nitrogen
   C. Sulfur
   D. Carbon

Explanation: **Legumes** are plants that harbour nitrogen fixing bacteria in their root nodules. Nitrogen fixing bacteria converts the atmospheric nitrogen gas (N_2) in to ammonium ions which later are converted to nitrates (the form plants absorb from these oil). When farmers include legumes in their crop rotation they improve the content of nitrogen compounds in the soil which is one of the important nutried for making protein in plants.

∴ ***Answer B***

6. Before making crosses, which part of the flower did Mendel remove to avoid self pollination?
   A .Stigma
   B .Ovule
   C. Ovary
   D. Stamens

Explanation: In his experiments of crosses Mendel crossed plants of contrasting characteristics (example, tall and short).In the crossing he transferred pollen grain from the stamens of one of the plant that donate pollen grain to the stigma of another plant the receive the pollen grain. Before doing this he removed the stamens of the plant that receive the pollen grain before they matured to avoid self pollination of this plant.

∴ ***Answer D***

7. Which one of the following do bees use to inform other bees about the location and distance of a new source of nectar they discover?
   A . Pheromones
   B .Waggle dance
   C. Buzzing noise
   D. Vibration of wings.

Explanation: **Workers bees** communicate with each other in a very special way to convey information about the location and distance of the new source of nectar. Foragers perform a **waggle- dance** on the honey comb to inform other workers of the direction of the nectar source and its distance. The dance takes the form of a **figure of eight** on the vertical face of the honey comb.

∴ ***Answer B***

8. One of the following molecules is the building units of an enzyme molecule.
   A .Amino acids
   B .Glucose
   C. Nucleotides
   D. Fatty acids

Explanation:

❖ **Amino acids** are the building blocks or units of protein. All enzymes are protein , hence the building blocks of enzymes.

❖ **Glucose** is the building blocks or units of **polysaccharides** such as ,starch glycogen and cellulose.

❖ **Nucleotides** are the building blocks or monomers of the nucleic acids (DNA and RNA) polymers.

❖ **Fatty acids** are parts of fats and oils. Three fatty acids join to one glycerol molecule forms a fat or oil molecule.

∴ ***Answer A***

9. Which one of the following is the main constituent of biological membranes?
   A .Phospholipids
   B .Glycoprotein's
   C. Glycolipids
   D. Cholesterols

Explanation:

❖ **Phospholipids** are formed when two fatty acid molecules are bonded to the glycerol and the place of third fatty acid is taken by a **phosphate group.** In water , phosphor lipids become organized in to bilayer. Phospholipids bilayers are the basis of **plasma membrane**

❖ **Glycoproteins** are protein with branching carbohydrate side chains like "antennae".

❖ **Glycolipids** also have branching carbohydrate side chains (lipids with branching carbohydrate side chain) and are involved in cell to cell recognition. They may act as receptor sites for chemical signals.

❖ **Cholesterols** are like unsaturated fatty acids cholesterol disturbs the close packing of phospholipids and keeps them more fluid. Cholesterol also increases flexibility and stability of membranes

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without which they break

∴ ***Answer A***

10. One of the following biomes in Africa is supporting large wild mammals such as elephants , giraffes and lions.
A .The Congo Rainforest
B .The Rain Forest of western Ethiopia.
C .The Savanna Grass land
D .The Sahara Desert.

Explanation:

- **TheCongo rain forest** is the part of tropical rain forest of the world having almost similar major characteristics. It is an ecosystem with high temperature and high rain fall. Epiphytes , climbers, evergreen and deciduous broad- leaved tree and bamboo are the vegetation of the ecosystem harbouring insects reptiles, birds and arboreal animals.
- **The rain forest of Western Ethiopia:** is distributed in Wollega, Illubabor, Bale and Southern Keffaat altitudes ranging from 800- 2500 m. The vegetation of this ecosystem is dominated by moist evergreen montane forest. The major mammals species are **blue monkey, Bush buck, Great forest hog, Wild pig ,** etc.
- **Savanna Grass lands** are tropical grass lands often with scattered trees. They are largely found in Africa, but also found in Australia, South America and Southern Asia. The savanna grass lands of East Africa are known for their wild games which are in abundance. These includes mammals such as elephants, rhinoceroses, leopards , giraffes and lions.
- The **sahara desert**: Deserts usually receive less than 250mm of annual , rain fall and rain is unpredictable. The animal life of the biome includes **rats ,insects and camels.**

∴ ***Answer C***

11. Which of the following is the correct F_2 phenotype ratio of a monohybrid cross ?
A .1 :2 B.1:1 C.3:1 D. 2 :2

Explanation: In monohybrid crosses when two plants one showing the dominant trait and the other, the recessive trait are crossed all of the offspring (F_1 generation)shows the dominant trait. If these F_1 offspring are selfed the off spring of these (F_2 generation) appears in the ratio of 3:1 .3plants with dominant trait to one plant with the recessive trait. Therefore, the phenotypic ratio of monohybrid cross is 3:1

∴ ***Answer C***

12. How many carbon atoms are there in one disaccharide molecule?
A .6 B. 12 C .18 D.24

Explanation: A **monosaccharide** molecule like glucose consists of 6 carbon atoms. A disaccharide like maltose which is composed of two monosaccharide (glucose) contains 12 carbon atoms.

∴ ***Answer B***

13. Which unit is best to use for measuring the smallest cells and organelles?
A .Micrometer C. Millimeter
B .Milliliter D. Nanometer

Explanation: We use smaller units to measure the size of cells and molecules. These smaller units gives us numbers that are more convenient to work with.

- Micrometer (µm) is $\frac{1}{1000}$ of millimeter or $\frac{1}{1{,}000{,}000}$ of meter.
- Millimeter is $\frac{1}{1000}$ of a meter.
- Milliliter is $\frac{1}{1000}$ of a liter.
- Nanometer is $\frac{1}{1000}$ of a micrometer or $\frac{1}{1{,}000{,}000}$ of a millimeter or $\frac{1}{1{,}000{,}000{,}000}$ of a meter.

∴ ***Answer D***

14. Which energy rich organic compound contains adenine in its molecule?
A .Lipid C. Glucose
B .Carbohydrate D.ATP

Explanation:

- **Lipids** are a varied group of compounds that include triglycerides . A tryglyceral (lipid) is composed of three fatty acids and a glycerol molecule.
- **Carbohydrates:** Contain elements carbon, hydrogen and oxygen. Carbohydrates are used to release energy in respiration .
- **Glucose** is a monosaccharide carbohydrate used to release energy in respiration .It is the primary source of energy of most organisms.
- **ATP (adenosine tri- phosphate)** is an energy storing compound. It is the immediate source of energy for all living cells. It is a molecule made of adenine , ribose sugar and three phosphate groups.

∴ ***Answer D***

15. What are the most frequent causative agents of food poisoning?
A . Bacteria C. Viruses
B . Protozoa D. Worms

Explanation:

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❖ **Bacteria :** most food poisoning is bacterial , but some viruses are transmitted this way. The organisms initially infect a region of the gut.
❖ **Protozoa** are unicellular eukaryotes. Some protozoa cause diseases like malaria and sleeping sickness transmitted by animal bites.
❖ **Viruses** enter living cells (they are all parasites)and disrupt the metabolic systems of the cell. Influenza ("flu"), AIDS , measles and common cold are some of the diseases caused by viruses.
❖ **Worms :**There are three classes of worms which parasitize human beings. They are nematodes or round worms, tape worms and trematodes or flat worms.

∴ ***Answer A***

16. What are the most likely causes of variations within species?
A . Mitosis and asexual reproduction.
B . Overpopulation and overproduction.
C .Vegetative propagation and cloning.
D .Mutations and sexual reproduction.

Explanation: The most likely causes of variation within species are mutations and sexual reproduction.

✓ **Mutation** are any spontaneous change in the genetic material of an organism. Mutation are important introducing variation in to populations . **Mutations** are the ultimate source of variation.
✓ In the sexual reproduction gametes are produced by meiotic cell division. **Meiosis** provides opportunities for new combination of alleles to occur in the gametes , which leads to genetic variation in the off spring by fusion of gametes. This genetic variation is achieved through independent assortment of chromosomes and crossing over in meiosis.In sexually reproducing organisms random fusion of gametes also provides genetic variation in a population

∴ ***Answer D***

17. In which hominid species did scientists find the smallest brain size (cranial capacity)?
A .Homo sapiens
B .Homo habilis
C. Homo erectus
D. Homo neanderthalensis.

Explanation:
❖ **Homo sapiens** have average brain size of about $1500cm^3$.
❖ **Homo habilis** have average brain size of about $1000cm^3$
❖ **Homo erectus** have a brain size of a little larger than $1000cm^3$.
❖ **Homo neanderthalensis** have a brain size near to that of Homo sapiens.

∴ ***Answer B***

18. Which of the following is NOT classified as a learned behavior?
A .Insight
B .Innate
C .Latent
D. Conditioned

Explanation:
❖ **Insight learning** involves finding solutions to problems that are not based on actual experience (as with trial and error) but on "trials" occurring mentally. Often the solution is learned suddenly such as when a person has been trying to solve a problem for a period of time and suddenly the solution appears almost "out of nowhere".
❖ The world **innate** literally means" in born". Innate behavior is behavior that is present (potentially) at birth or hatching. It does not have to be learned.
❖ **Latent :**The word "latent "means "**hidden**" so we are taking about "**hidden learning** "or learning that is not apparent as it takes place. Latent learning happens when the brain acquires knowledge at a certain time, without reinforcement, but does not use it until later, at a time when that knowledge is needed.
❖ **Conditioned learning :** is used to get an animal to learn a particular behavior by associating that behavior with event or stimulus or with a consequence such as reward or punishment.

∴ ***Answer B***

19. What is the specialist in biology called if he/she studies fossils to generate new knowledge on the origin and evolution of living things of past geologic periods?
A .Geneticist
B .Ecologist
C. Paleontologist
D. Comparative Biochemist.

Explanation:
❖ **Geneticist** is a biologist that study heredity and variation and their physical basis in DNA ,the genetic material of an organism, the physical basis of its inherited characteristics.
❖ **Ecologist** is a biologist that study the interrelation ships between organisms and their environment and each other.
❖ **Paleontologist** is a biologist who try to find out more about the way in which life began on earth and how it has evolved from simple forms in to more complex ones.
❖ **Comparative biochemist** is a biologist that study chemical reactions occurring in living organisms

∴ ***Answer C***

20. All proteins contain carbon, hydrogen , oxygen and what other element?
A .Chlorine
B .Fluorine
C. Nitrogen
D. Sulphur.

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Explanation: All **proteins** contain the elements carbon, hydrogen oxygen and nitrogen. Nitrogen makes the amino group of the amino acids that make proteins.

∴ ***Answer C***

21. What makes unsaturated fatty acids different from saturated fatty acids?
A . The presence of long chain of carbon.
B . The presence of one or more double bonds.
C . The presence of large number of hydrogen atoms.
D . Their occurrences as solid at room temperature.

Explanation: **Fatty acids** sometimes contains one or more double bonds (c = c) , such as oleic acid. In this case they are said to be unsaturated, as are lipids containing them. If one of the carbon- carbon bond is double bond, then it is a monosaturated fatty acid. If more than one carbon- carbon bond is a double bond, then the fatty acids is a poly unsaturated fatty acid.

✓ **Fatty acids** lacking double bonds between carbon atoms are said to be saturated. Unsaturated fatty acids melt at lower temperature than saturated fatty acids.

∴ ***Answer B***

22. Which of the following pairs of molecules are known to have catalytic activity?
A .Lipids and proteins. C. Proteins and RNAs
B .Carbohydrates and proteins. D. Proteins and DNAs

Explanation: **Lipids** are sometimes classified loosely as those water-insoluble organic substance than can be extracted from cells by organic solvents such as ether, chloroform and benzene. They are composed of carbon, hydrogen and oxygen. They differ from carbohydrates in that they contain much smaller proportion of oxygen. The major function of lipids is to act as energy store.

✓ **Carbohydrates** are substances which contain the elements carbon, hydrogen and oxygen and have the general formula $C_X(H_2O)_Y$. **Proteins** are far more complex than either carbohydrates or lipids. **proteins** function in vision , in oxygen transport , in the immune response and as hormones. They are also catalysts of biological reactions as enzymes.

✓ A second important category of nucleic acid comprises the ribonucleic acids, or RNA. There are several types of RNA ,each with different role in protein synthesis such as messenger RNA (m RNA), ribosomal RNA (r RNA) and transfer RNA (t RNA).

∴ ***Answer C***

23. If a suspension of a mixture of cellular organelles is spun in a centrifuge, which organelle settles to the bottom first?
A .Mitochondria C. Chloroplasts
B .Nuclei D. Ribosomes.

Explanation: **Biologists** have been able to study the different organelles of cells because of the presence of a techniques called **cell fractionation**. The techniques is based on the fact that the masses of organelles vary and depends on their size. When a mixture of organelles is spun in centrifuge, the various types settle out at different speed of spinning. The large nucleus requires relatively low centrifuge speed to make it settle out ; the much smaller ribosome's require a much higher speed. There for nuclei are the first to settle to the bottom and **ribosome's** are the last.

∴ ***Answer B***

24. During chemiosmosis, what substance diffuses from one side to the other side of the membrane?
A .Water molecules C. Electrons
B .Protons D.ATP molecules

Explanation: **In electron transport chain** , electrons pass along a series of electron carriers that form the transport chain; they lose energy as they pass from one carrier to the next. Three of these electron carriers are proton pumps that move protons from the matrix of the mitochondrion to the intermembrane space (space between the inner membrane and outer membrane).

∴ ***Answer B***

25. Which of the following human diseases can be prevented by taking proper diets?
A .Degenerative diseases. C. Social diseases.
B .Genetic diseases. D. Deficiency diseases

Explanation:

- **Degenerative diseases** often result from aging process during which the affected tissues. Deteriorate over time due to simple " Wear and tear "
  Example: Artheritis and artherosclerosis.
- **Genetic diseases** are diseases that result from the action of mutated genes.
  Example: Hemophilia and sickle cell anemia.
- **Social diseases** are conditions that result from social activities and may lead to socially unacceptable behavior such as alcoholism and drug addiction
- **Deficiency diseases** are disease that result from a lack of a nutrient in our diet such as **scurvy** (caused by lack of **Vitamins C** and **Kwashiorkor** (caused by lack of **protein**).

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∴ *Answer D*

26. When two species are compared , which of the following sources of evidence is least informative about the degree of relationships between the species?
A .Nucleotide sequence of DNAs.
B .Amino acid sequences of proteins.
C .Glucose sequence of polysaccharides.
D .DNA – DNA hybridization.
Explanation:
❖ **Nucleotide sequence** or similarities in base sequence of DNAs of different species suggests the evolutionary relationships between the species or their common ancestral origin.
❖ The similarity in the amino acid sequence of proteins such as hemoglobin and cytochrome of different species shows the evolutionary relationships between the species or their common ancestral origin.
❖ As all polysaccharides are made from the same monomers of glucose ($\propto$ glucose or β - glucose)there is no difference in the sequence of glucose of polysaccharides of different species. Therefore , this is not used as evidence suggesting evolutionary relationship between different species.
✓ **DNA – DNA hybridization** is a technique used to find out the extent of binding of one strand of DNA with another strand of another DNA. The similarities of base sequences between different DNA is investigated using this technique.

∴ *Answer C*

27.Which of the following practices does **NOT** normally transmit HIV ?
A. Sexual intercourse through anus.
B . Sharing injection needles.
C . Blood transfusion.
D . Shaking hands.
Explanation: **Sexual intercourse** between the same sex or different sex, sharing injection needles which results blood to blood contact and blood transfusion which mixes up blood of different individuals are the major way of transmission of HIV infection. But shaking hands does not result blood – to – blood contact and therefore it is not a means of transmission of the HIV infection.

∴ *Answer D*

28.What is the important role played by microorganisms such as bacteria and fungi in the ecosystem?
A .Antibiotic production.
B .Recycling of nutrients.
C .Forming organic substances.

D .Supplying energy to the ecosystem
Explanation: Bacteria and fungi are the principal agents of decomposition and are collectively known as decomposers. Therefore, micro-organisms play the most important role of recycling nutrients in the ecosystem.

∴ *Answer B*

29. In which one of the following aspects is the tropical rainforest biome poor?
A .Species diversity
B . Amount of sunlight
C. Annual precipitation
D. Soil fertility.
Explanation: **Tropical rain forest** is one of the most complex and species – rich (highest species diversity) ecosystem in the world. It is a very warm ecosystem receiving the highest amount of rain fall. Although many of the trees are **very tall, the root systems are shallow** and trees can easily fall. The shallow root systems grow in shallow, **nutrient- poor soil**. The soils are nutrient poor because many of the minerals from the soil remain **"locked up "** in the huge trees.

∴ *Answer D*

30. If the size of a cell increases, which one of the following gets smaller?
A .The volume of the cell
B .The surface area of the cell.
C .Surface area to volume ratio of the cell.
D .Volume to surface area ratio of the cell.
Explanation: As the cell increases in size , the volume increases faster than the surface area , meaning the surface area decreases. The surface area of the cell is the area through which the cell makes contact with the environment. The surface area to volume ratio of the cell is its total surface area divide by its volume. The bigger the volume the less the surface area to volume ratio. This affects the movement of materials in to and out of the cells. Cells with higher surface area to volume ratio are more efficient in the transportation of substance in to and out of the cell than cells of lower surface area to volume ratio.
Volume to surface area ratio of the cell is not considered usually as important factor in exchange of materials.

∴ *Answer C*

31.Which of the following is **NOT** usually true as an ecological succession progresses to advanced seral stages?
A .More ecological niches are formed.
B .Species become more diverse.
C .The depth of the soil increases.
D .Less populations are supported.

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Explanation: **Succession** is a progressive series of change in vegetation and animal life of an area from initial colonization to final stabilizing stage, or climax.
The following trends occur as an ecological succession progresses to advanced serial stages:

- the abiotic conditions (physical conditions)change.
- the changed abiotic conditions allow other species to colonize the area.
- the total biomass of the community increases.
- the number of ecological niches increases.
- food web become more complex.
- the community become more stable- Climax community.

∴ ***Answer D***

32. One of the following is an important cause of gene mutation.
A .Old age C. Smoking
B .Alcoholic drinks D. Radiation

Explanation: **Mutations** can be either spontaneous or induced.
Spontaneous mutations are those having no known cause to occur.
Induced mutations on the other hand, can be caused by accidental or deliberate exposure of organisms to mutagenic (mutation causing) agents such as high energy **electromagnetic radiation** like ultra violet rays and x- rays, radioactive particles and high energy particles such as neutrons, cosmic radiation as well as some artificial chemicals such as mustarded gas , formaldehyde , certain constituents of tobacco etc.

∴ ***Answer D***

33. What is the reproductive isolating mechanism called if two species of frogs do not interbreed because they cannot understand the mating calls of one another?
A .Seasonal isolation C. Temporal isolation
B .Behavioral isolation D. Isolation by distance.

Explanation:

- **Seasonal isolation** occurs when members of the two populations reproduce at different time of the year.
- **Behavioral isolation** occurs when members of the two populations have different court ship patterns.
  Court ship behavior is an activity that proceeds and result in mating and reproduction. It allows members of a species to recognize each other and prevents or reduces attempts at inter breeding between different species.
- **Temporal isolation** occurs when members of the two populations reproduce at different times of the day.
- **Isolation by distance** refers to isolation by some physical features such as:

- a rivers changing the course.
- a mountain range being created.
- a land mass separating two bodies of water.
- these type of isolations are known as **geographical isolation.**

∴ ***Answer B***

34. How many chromosomes do humans inherit from each of their parents?
A .23 chromosomes C.46 chromosomes
B .23 pairs of chromosomes D. 46 pairs of chromosomes.

Explanation: The human body cells somatic cells contain 46 chromosomes (diploid chromosomes number).

✓ In the process of gamete formation, meiosis, the diploid cells(46 chromosomes) are reduced in to haploid cells in both male and female. Because meiosis is a reduction division.

✓ The male parents produce gametes (sperm cells) containing 23 chromosomes and the female parents similarly produce gametes (egg cells) containing 23 chromosomes.

✓ Humans are produced from a zygote formed by the fusion of sperm cells containing 23 chromosomes and egg cells contain 23 chromosomes each.

∴ ***Answer A***

35. In which one of the four phases of population growth is the number of the population the highest?
A .Lag phase C. Constant phase
B .Log phase D. Decline phase.

Explanation:

- **Lag phase** is a growth phase where a population establishes itself; in the process some organisms are not adapted to the environment and die, others reproduce. The numbers remain low and static or increases slowly.
- **Log phase:** all members of the population are adapted and reproduce rapidly due to plentiful resources.
- **Constant phase (Stationary phase)** the carrying capacity of the area is reached the same number are dying as produced in reproduction. Numbers remain fairly constant ; they fluctuate around the "mean" level.
- **Decline phase** nutrients are exhausted, a new disease strikes or toxic execratory products accumulate. Numbers decline rapidly.

∴ ***Answer B***

36. Which of the following is **NOT** true about the gene called SRY?
A. It is found on the Y- chromosome.
B .It determines maleness.
C .Testes develop in its presence.
D .Females have two copies of this gene.

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Explanation: It appears that Y- chromosome determines a person's sex. It is , in fact , the action of one gene on this chromosome- the SRY gene – that determines the formation of tests.

✓ In early development of the embryo, a region called the urogenital ridge develops in to a "bi-potential gonad". This means that the same structure can develop in to either ovary or testis.

✓ When SRY gene is activated , the bipotential gonad develops in to testis, and the embryo is male. If the SRY gene is absent (or inactivated for some reason)genes on the x –Chromosomes and on the autosomes cause the bi-potential gonad to develop in to an ovary, and the embryo is female.

∴ ***Answer D***

37. To which of the following groups do those bacteriophages that integrate their DNA into the chromosome of their bacteria host belong?
A. Virulent viruses
B . Lytic viruses
C. Lysogenic viruses
D. Non- parasitic viruses .

Explanation: **Virulent viruses:** Bacterio phages that kill their host cells are said to be virulant and their type of replication cycle is known as **the lytic cycle.**

✓ **Lytic viruses** are viruses the reproduce by lytic cycle.

✓ lysogenic viruses are viruses that reproduces by Lysogenic cycle. In this type of viral reproduction, the viral DNA becomes incorporated in to the DNA of the host cell. Each time the cell divides, the DNA is replicated and each daughter cell gets a copy of the cell DNA , which now includes the viral DNA.

∴ ***Answer C***

38. What do we call the learned behavior if a mouse that had just escaped from the mouth of a cat jumped violently at a slight touch by a trivial object?
A .Latent learning
B .Sensitization
C .Conditioning
D .Imprinting

Explanation:

❖ **Latent learning or hidden learning** is a learning that is not apparent as it takes place. Latent learning happens when the brain a quires knowledge at a certain time, without reinforcement, but does not use it until later at a time when the knowledge is needed

❖ **Sensitization** is a type of learning where there is an increase in the response to a harmless, stimulus when the stimulus occurs after a harmful stimulue. A person encountered to a snake in an isolated condition, jumps violently when exposed to anything long and slender like a piece of rope.

❖ **Conditioning** is the animal learns a particular behavior by associating that behavior with event or stimulus or with a consequence such as reward or punishment.

❖ **Imprinting** is a kind of instinctive behavior in which the fixed action pattern is for newly born /hatched organisms to imprint on (or become attached to) the first thing they see that has certain general features (those of an adult of its species.)

∴ ***Answer B***

39. Which of the following theories explains evolutionary changes of living things in terms of changes in their allele frequencies?
A .Darwin's natural selection.
B .Spontaneous generation.
C .Inheritance of acquired characteristics.
D .Neo-Darwinism.

Explanation: **Neo- Darwinism** is the modern aspect of Darwinism that incorporates the knowledge of gene and gene action in to the theory of natural selection to give a better understanding of what drives evolution. According to this modern view of natural selection:

- those individuals with the advantageous allele of the gene will survive to reproduce in greater numbers than other types.
- they will pass on their advantageous allele in greater numbers than the other types pass on their allele of the same gene.
- the frequency of the advantageous allele in the gene pool of the population will be higher in the next generation.
- this process 'repeats' over many generation and the frequency of the advantageous allele in the gene pool increases with each generation that passes

∴ ***Answer D***

40. Which of the following is an important function of the Golgi apparatus?
A .Protein synthesis
B .Packaging of proteins for export out of the cell.
C .Removing of debris from cell.
D .Storage of waste materials not needed by the cell.

Explanation:

❖ **Protein** are synthesized on the **ribosome's** which are also known as protein factories of a cell.

❖ **The Golgi apparatus** consists of a number of flattened membrane bounds sacs, in which proteins are modified .Many of the modification added in the Golgi apparatus acts as a kind of **"tag"** which determine the final destination of the molecules. Think of the Golgi apparatus as a cellular post office that labels and then distribute molecules. Many of the modified molecules are released from the Golgi apparatus in

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vesicles to be carried to other parts of the cell or to the plasma membrane to pass out of the cell by exocytosis to be used elsewhere.

∴ ***Answer B***

41. Which of the following field equipment is used to determine an exact location of a place?
A . Theodolite
B .Global positioning system
C. Field microscope
D. Field PH kit

Explanation:

- **Theodolite** is a field tool used by biologists to measure the height of trees or slopes in the area.
- **Global poisoning system (GPS) :** This instrument makes it possible to record positions quickly and extremely accurately. By taking several readings at different points on the perimeter of the area, an accurate map of the area can be drawn.
- **A field microscope** is used to investigate the structure of specimens in the field, while still fresh.
- **Field PH kit** is used to measure the PH of soil or water.

∴ ***Answer B***

42. The sheep "dolly" is an example of which biotechnological manipulation of animals by human?
A .Transgenic animal.
B .Genetically engineered animal
C. Cloned animal
D. Hybrid animal

Explanation:

- **Transgenic animals** also known as genetically modified animals are animals whose genome (genetic makeup) is altered by transferring gene from another organism in to their DNA by a new biotechnology called **genetic engineering.**
- **Genetically engineered animals** are transgenetic animals
- The term **clone** is often applied to whole organisms as well as to genes. A clone of organisms is a group of organisms produced asexually from one parent. The members of the clones are genetically identical to each other and to the parent organism.
- A cloned animal like Dolly the sheep was produced by transferring a diploid nucleus to an egg cell that had enucleated (had its nucleus removed). **A hybrid animals** is an animal that is produced from two different breeds of animals of the same species by a selection known as **selection breeding.**

∴ ***Answer C***

43. Among the following , which one is the best criterion to show that two populations belong to same species?
A .Morphological similarity
B .Physiologically similarity

C .Inhabiting the same geographic area.
D .Production of fertile off spring.

Explanation: A **species** is a group of similar ,inter breeding organisms that produce fertile off spring. If two population become so different that individuals from different populations cannot inter breed to produce fertile offspring then we must think of them as different species.

✓ When two populations inter breed and produce a fertile offspring we say that the two populations belong to the same species.
This is a technique used by biologists to find out whether or not two individuals of unknown species belong to the same species.

∴ ***Answer D***

44. In the process of the evolution of life on earth, which of the following four processes evolved last?
A .Photosynthesis
B .Aerobic respiration
C. Chemosynthesis
D. Photo – autotrophism

Explanation: The fossil record shows that cynobacteria had been producing oxygen by photosynthesis from about 3.5 billion years ago.

✓ **Photo-autotrophy** are organisms that use sunlight energy to make organic compounds by the process of photosynthesis. The origin of photo-autotrophy means the origin of **photosynthesis.**

✓ The oxygen produced by the photoautotroph's had made it possible for **aerobic respiration** to evolve as an energy releasing path way. This happened about 600 to 700 million years ago.

✓ Scientists believe that the same of first organisms to inhabit Earth were chemo-autotrophs. **Chemo-autotrophs** are organisms that chemo synthesize (manufacture) organic compounds using energy from chemical reactions as a source of energy.

∴ ***Answer B***

45. What is a theory in biology?
A .The outcome of an experiment.
B .A proof that shows the hypothesis is true.
C .A hypothesis supported by experiments and / or observations.
D .An opinion or educated guess resulting from observation.

Explanation: When a given hypothesis has been tried (tested) many times and found to be acceptable it is no longer called a **hypothesis.** It is automatically transformed in to a theory. When a scientist signifies a statement by the name theory, it means that it has a very high degree of probability with a great degree of confidence. A theory is then a hypothesis that has been tested to be true.

∴ ***Answer C***

46. For which of the following is a theodolite used in biology education?
A .Measuring the height of trees.

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B .Recording positions where a species is found.
C .Measuring the rate of flow of water in a cell.
D .Measuring the PH of water or soil.

Explanation: **A theodolite** is a tool used by biologists in the field to measure the height of trees or slopes in an area.

∴ ***Answer A***

47. What is the name of the sugar found in milk?
A .Glucose
B .Lactose
C. Maltose
D. Sucrose.

Explanation:

- **Glucose** is a monosaccharide ($C_6H_{12}O_6$) occurring widely in nature. It occurs as glucose units in glycogen, sucrose, starch and cellulose.
- **Lactose (milk sugar)**is a disaccharide derived from a β- glucose molecule and an $\propto$ - galactose molecules. It is found exclusively in milk and it is important energy source for young mammals.
- **Maltose** is a disaccharide that mainly occurs as a break down product during digestion of starch by enzyme called **amylase.**
- **Sucrose ,or cane sugar,** is the most abundant disaccharide in nature. It is most commonly found in plants.

∴ ***Answer B***

48. To which class of enzymes do the digestive enzymes belong?
A .Esterase's
B .Transferases
C. Hydrolases
D. Isomerases

Explanation:

- **Esterases** are enzymes that breaks bonds formed between the carboxyl group and hydroxyl group known as **Ester bond.**
- **Transferases**: catalyze the transfer of a chemical group from one substrate to another.
  Example : phosphotransferases – catalyze the transfer of a phosphate group from one molecule to another.
- **Hydrolases:** Catalyses hydrolysis reactions, i.e the breaking down of large molecules by the addition of water.
  Example : digestive enzymes such as lipases that break down lipids in to fatty acids and glycerol. Digestive enzymes are known **as hydrolytic enzymes.**
- **Isomerases**: Catalyze internal rearrangement of substrate molecules (i.e isomerisation)
  Example :phosphor hexo isomerase rearranges glucose – 6 – phosphate to fructose-6-phosphate.

∴ ***Answer C***

49. In which of the following features are eukaryotic cells distinguished from prokaryotic cells?
A They have mitochondria.
B .Their nuclei lack membranes.
C .They have no DNA.
D. They have smaller ribosome's

Explanation: All cellular organisms fall in to two natural groups, known as **prokaryotes and eukaryotes.**
In prokaryotes the DNA is not enclosed by nuclear membranes and lies free in the cytoplasm. These cells therefore lack true nuclei.
In eukaryotes, the DNA is enclosed by a nuclear membrane and they contain true nuclei. In addition to this eukaryotes have several membrane bounded organelles such as **mitochondria, chloroplast, lysosomes and Golgi apparatuses.** There are no such organelles in prokaryotes.

∴ ***Answer A***

50. What will happen if human red blood cells are kept in a hypotonic solution?
A Lose water by osmosis and burst.
B .Lose water by osmosis and shrink.
C .Take in water by osmosis, swell and burst.
D .Take in water by osmosis, swell and remain turgid.

Explanation: **A hypotonic solution** is a solution whose solute content (higher water potential)is less than the cell solution (lower water potential).Since water moves from its higher water potential to a region of lower potential in red blood cells that are kept in hypotonic solution, water moves in to the cells by osmosis and the cells **swells and burst.**

∴ ***Answer C***

51. In which type of solution is the water potential more negative than in the cells?
A. Hypotonic
B .Hypertonic
C. Isotonic
D. Equal solute and solvent concentration.

Explanation:

- **Hypotonic:** If the concentration of the solution on the outside of the cell membrane is lower than the concentration of the solution inside the cell, it is **hypotonic.** If plant cells are placed in such a solution water moves in to the cell making the cell **turgid.** Hypotonic solutions have more water potential than the cell solution. But still have negative water potential.

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- **Hypertonic solution:** If the concentration of the solution on the outside of the cell membrane is higher than the concentration of the solution inside it is **hyper tonic**. If plant cell are placed in such a solution water moves out of the cells in to the solution resulting shrinkage of the cells (loss of turgor pressure, therefore turgidity). **Hypertonic solutions** have lower water potential than the cell solution and hence the maximum negative water potential.
- **Isotonic solution** : If the concentration of the solution on both side of a cell membrane are the same they are **Isotonic.** If plant cells are kept in such solution water neither moves out of the cell nor in to the cell. **Isotonic solutions** have the same water potential as the cell solution. They have more negative water potential than hypotonic solution and less negative water potential than hypertonic solutions.
- **Solutions** with equals solutes and solvent concentration are isotonic solution.

∴ ***Answer B***

52. Which of the following paired organelles are membrane- bound?
A .Ribosome's and peroxisomes.
B .Chloroplasts and ribosome's.
C .Mitochondria and ribosome's.
D .Chloroplasts and mitochondria.

Explanation:

- **Ribosomes** are not membrane bounded. They are the site of protein synthesis made from ribosomal RNA and protein.
- ✓ **Peroxisomes** are single membrane bounded organelles, that are concerned with glycolate metabolism inphotorespiration.
- **Chloroplast** are double membrane bounded organelles and are the sites of photosynthesis.
- **Mitochondria** are double membrane bounded organelles. They are the sites of cellular respiration.

∴ ***Answer D***

53. Most cell membranes are primarily composed of which compounds?
A .Proteins and lipids. C. Chitin and starch
B .DNA and ATP D. Nucleotides and amino acids

Explanation: Most cell membrane are primarily composed of :

- Phospho lipid bi layer that make the bulk of the membrane and
- **Protein** integrated in the phospho lipid bilayer.

∴ ***Answer A***

54 .Which group of micro – organisms causes the disease known as athlete's foot?
A .Bacteria C. Protozoa
B .Fungi D. Viruses

Explanation:

- ***Bacteria*** cause diseases such as pneumonia, cholera, pulmonary tuber closes, typhoid fever etc.
- ***Fungi*** causes – Athlete's foot, farmer's lung etc.
- ***Protozoa*** cause diseases such as malaria and sleeping sickness.
- ***Viruses*** cause – diseases such as Influenza, AIDS, measles and common cold.

∴ ***Answer B***

55. Choose the one that is different from all the others.
A .Genetically modified organisms.
B .Genetically engineered organisms.
C .Pathogenic organisms.
D .Transgenic organisms.

Explanation: **Genetically modified organisms** , genetically engineered organisms and transgenic organisms all the phrases refer to the same organisms whose genome (genetic makeup) is altered by adding or introducing new genes in to their DNA.

- **Pathogenic organisms** are organisms that cause disease.

∴ ***Answer C***

56. Under which of the following groups can the fungi be more conveniently placed?
A .Autotrophy. C. Prokaryotes
B .Heterotrophs. D. Plants.

Explanation: **Fungi** do not have true roots, stems or leaves. They are neither plants nor animals. Fungi do not make their own food (are not autotrophs) but they are heterotrophs that live as saprophytes and some as parasites.

∴ ***Answer B***

57. What is the type of community called when it has reached the final and most complex stage of a succession?
A .Pioneer community C. Climax community
B .Seral community D. Secondary community

Explanation:

- **Pioneer community** is the first group of organisms that colonized the uncolonized area in primary succession in the process of formation an ecosystem.
- **Seral community** is a community that is found at every stage of the ecological succession.
- **Climax community** is the final, most complex, state of a succession.
- **Secondary community** is a community that follows the pioneer community.

∴ ***Answer C***

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58. Which of the following is an ecosystem?
A .A tropical Rainforest.
B .The Africa continent
C .All the organisms in a given area.
D .The non- living components of an environment.
***Explanation* : Ecosystem** is a unit made up of all the living and non-living components of a particular area that interact and exchange materials with each other.
✓ **Tropical rain forest** is an ecosystem with its unique living and non-living components of a particular area specifically around the equator of the earth.

∴ ***Answer A***

59. Which of the following crops is considered to be the best choice for a better balance of essential amino acids as a human diet?
A .Maize C. Rice
B .Quinoa D. Wheat.
***Explanation* :**The best non- animal sources of amino acids are **quinoa, buck wheat, hempseed and amaranth,** although these contain lower overall amounts of proteins than some cereals (wheat , rice , maize)and nuts and pulses.

∴ ***Answer* B**

60. What is the main reason for the high species richness of plants and mammals observed in Ethiopia?
A .Lack of predators
B .Lack of disturbance
C .Presence of several biomes with in the country.
D . Efficient management of the ecological resources.
Explanation: Ethiopia is an important regional center of biological diversity. The wide range in altitude and climate, the isolation of the highlands of Ethiopia and the fact that there are so many different biomes present in the country, are some of the reasons for its high biodiversity (species richness).

∴ ***Answer C***

61. In which of the following are flowers and fruits found?
A .Ferns and relatives C. Mosses and conifers
B .Gymnosperms and ferns D. Monocots and dicots.
Explanation:
- **Ferns and relative plants** are vascular plants (with xylem and phloem)but are non- flowering plants meaning with no fruits and seeds.
- **Gymnosperms** are vascular plants with no flowers. They reproduce sexually by structure known as cones and produce **necked seeds**. Seeds that are not enclosed by fruits.

- **Mosses** are non-vascular plants. They have no true roots, stems and leaves, flowers. Mosses do not produce seeds and fruits.
- **Monocots and dicots** are vascular flowering plants producing fruits and seeds by sexual reproduction.

∴ ***Answer D***

62. In a cross between heterozygote's what proportion is expected to be homozygous recessive?
A .25% B. 50% C. 75% D. 100%
Explanation: In monohybrid crosses, a cross between two heterozygote's (individuals with one dominant allele and one recessive allele)always produce off spring in the ratio of 3:1 .3 off spring showing the dominant trait to one showing the homozygous recessive traitor this can be expressed in different way that is ¾ of the off spring with the dominant trait 75% and trait ¼ of the off spring with homozygous recessive trait (25%).

∴ ***Answer A***

63. Gene silencing is the function of one of the following molecules.
A . ds RNA C.si RNA
B . m RNA D. t RNA
Explanation: One group of substance that can act to repress gene action is known **as short interfering RNA (si RNA)** .These RNA molecules are unusual because they are very short- only about 21 to 23 nucleotides long- and are double stranded. They do not act on gene itself, but they interfere with or silence the **mRNA (messenger RNA)** a molecules that carries code messages from the gene – DNA)once it has been transcribed from the DNA. This is called **post transcription interference.**
Biologist think that the action of si RNA is as follows:
- Double – stranded RNA **(ds RNA)** is produced in the nucleus from a range of genes.
- It is then split in to the very short lengths that characterize Si RNA by an enzyme called **"Dicer".**
- The antisense strand of the si RNA then binds with a complex of molecules called RISC (RNA Induced silencing complex).
- The **Si RNA** binds with m RNA allows RISC to degrade or cleave the m RNA in to small fragments.

∴ ***Answer C***

64. Which process is held responsible for chronic myelogenous leukemia?
A .Translocation C. Transcription.
B .Translation D. Duplication..
Explanation:

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- **Translocation** is a type of chromosomal mutation where a piece of one chromosomes is transferred to another non- homologous chromosome. This type chromosomal mutation is often responsible for **chronic myelogenous leukemia.**
- **Translation** is a process in the protein synthesis where code messeges on the m RNA (messenger RNA) are translated to amino acids that join to form a polypeptide chain (protein).
- **Transcription** is the DNA code for the protein to be synthesized is rewritten in a molecule messenger RNA (m RNA) .
- **Duplication** is the replication of DNA molecule where it forms its own exact copy. It is a vital process in both mitotic and meiotic cell division.

∴ ***Answer A***

65. Two parents of genotype Aa are cross- bred. The alleles show complete dominance. What proportion of the offspring will phenotypically look like their parents?

A .0 B. $\frac{1}{4}$ C. $\frac{1}{2}$ D. $\frac{3}{4}$

Explanation: Parental genotype : A a × A a

Gamete formation or meiosis.....

Gametes..... A a A a

Fertilization

	A	a
A	AA	Aa
a	Aa	aa

Phenotype $\frac{3}{4}$ of the off spring show the **dominant trait** that is the parental phenotype $\frac{1}{4}$ of the off spring show the **recessive trait.**

∴ ***Answer D***

66. Which one of the following concepts contains all the others?

A Species C. Population

B .Genus D. Community

Explanation:

- **Species :** a similar group of organisms , all the members of which are able to breed amongst themselves and produce fertile offspring.
- **Genus:** in biology, category of classification of living things; specifically ,a group of species closely related in structure and evolutionary origin.
- **Population:** a group of organisms of the same species occupying a particular area at the same time.
- **Community:** a general term couering any naturally occurring group of different organisms living together in a certain environment, and interacting with each other.

∴ ***Answer D***

67. Choose the one that is **NOT** a requirement of all living things.

A .Ability to think.

B .Organization of parts.

C .Response to the stimuli.

D .Maintenance of internal constancy.

Explanation: **Ability to thinking** is not the characteristics of all organisms. These is the characteristics of higher animals like humans with complex and highly developed nervous system.

✓ **Organization of parts, response to stimuli and maintenance of internal consistency (homeostasis)** are the common characteristics of all organisms vital to their survival.

∴ ***Answer A***

68. What is the advantage of using HAART (highly active anti- retroviral therapy) for the treatment of HIV?

A .It gives a lasting immunity to HIV.

B .It prevents mutation of HIV.

C .It prevents re- infection by HIV.

D .It helps to break the life cycle of HIV.

Explanation: Although there is no cure for AIDS and , as yet no vaccine to give immunity against infection there are a number of drugs- called **anti- retroviral drugs** that can be effective in slowing down the progress to AIDS. These drugs work by blocking the reproduction of the virus in the CD_4 lymphocytes. There are several different drugs that act in different ways at different stages of the cycle of reproduction. Because the drugs act on different stages of the HIV life cycle, the most effective treatment is obtained when they are used together. This is called **high activity Anti- retroviral treatment (HAART).**

∴ ***Answer D***

69. Which of the following characteristics can show the evolutionary relationships among organisms?

A .Structures having similar functions.

B .Structures having same size.

C .Structures having common origin.

D .Structures having different origins.

Explanation: Some organisms have anatomical structures that are very similar in form and common evolutionary origin, but very different in function. We call such structures homologous structures. Because they are similar , they indicate an evolutionary relationships and common ancestor of the species that posses them.

∴ ***Answer C***

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70. The group of the Ethiopian wolf which does **NOT** contribute to territory marking with their urine containing pheromones is _____.
A ,Adult males
B .Adult females
C. Sub- adult males
D. Sub – adult females.

Explanation: The Ethiopian wolf is a social animal; the wolves live and hunt in packs. As a result, they maintain a group territory by making with urine (containing pheromones).All adult animals (male and female) contribute to this making behavior, particularly during patrols of the territory. Some of the sub adult (younger) males occasionally mark but sub adult females never mark.

∴ ***Answer D***

71. What can one conclude about the contents of the food stuff if a mixture of Benedict solution and a food stuff remain blue after heating?
A .It contains proteins
B .It lacks starch
C. It contains fats.
D. It lacks reducing sugar.

Explanation: The presence or absence of reducing sugars in a food stuff is tested using the **benedicts' test** for reducing sugars.
Reducing sugars include glucose, fructose, maltose and lactose. They reduce the blue Benedict's solution in to a brick red solution. When a mixture of food stuff and Benedicts solution is heated in a water bath for five minutes, brick red precipitate (or yellow ,orange, red colour) is formed. This indicates the presence of reducing sugar. If no colour change is observed it shows that the food stuff does not contain reducing sugars.

∴ ***Answer D***

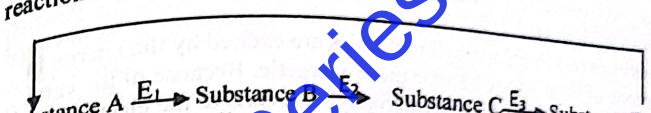
72. Which of the following types of enzyme inhibitions can be removed when the end product of the metabolic path way is depleted?
A . Allosteric inhibition.
B .Non- reversible inhibition.
C .Competitive inhibition.
D .Reversible competitive inhibition.

Explanation: Many substances are produced in cells as a result of a metabolic path way (a series of reactions), which can be represented as:

A substance $\xrightarrow{E_1}$ Substance B $\xrightarrow{E_2}$ Substance C $\xrightarrow{E_3}$ Substance D

E_1 , E_2 ,and E_3 are enzymes catalyzing the reactions. All the reactions in this sequence are enzyme controlled. Therefore, inhibition of any of these enzymes will interrupt the process .However, the main function of this path way is to produce substance D for use by the cell. If the requirement for substance D in the cell decreases, then the concentration of D will increase. This is least inefficient and may be potentially harmful because high concentration could be toxic .Such reaction sequences are often controlled by end – product inhibition .The end

product (D) inhibits the enzyme controlling the first stage of the reaction sequences as shown in the diagram.



Substance D normally acts as a non- competitive inhibitor, which prevents the enzyme E1 from catalyzing the reaction that converts A to B . As a result the entire reaction sequence is halted. When the inhibition is removed (the non- competitive inhibitor D), the metabolic path way can start again. Sometimes, a decrease in the concentration of D is sufficient to remove the inhibition. Non - competitive inhibitors are allosteric inhibitors.

∴ ***Answer A***

73. If red blood cells shrink when placed in a certain solution, what is the strength of the solution relative to the strength of the protoplasm of the cells?
A .Hypotonic
B .Isotonic
C. Hypertonic
D. Is osmotic.

Explanation:

- **Hypotonic:** If the concentration of the solution on the outside of the cell membrane is lower than the concentration of the solution inside the cell's protoplasm. It is hypotonic. If red blood cells are placed in such solutions, water moves in to the cell making the cells swell and burst.
- **Isotonic solution**: If the concentration of the solution on both sides of a cell membrane are the same they are isotonic. If red blood cells are kept in such solution water neither moves out of the cell nor in to the cell. The cells remain the same.
- **Hypertonic solution**: If the concentration of the solution on the outside of the cell membrane is higher than the concentration of the solution inside, it is hypertonic. If red blood cells are kept in such solutions, water moves out of the cells and then, the cells shrink as a result of losing water.
- **Is osmotic** refers to the similarity of two solutions in their osmotic potential

∴ ***Answer C***

74. Which of the following is **NOT** true about photo system – II ?
A . Its reaction center molecule is p680.
B .It passes its excited electrons to photo system – I.
C .The energy lost from its excited electrons reduces NADP.
D .It replenishes its lost electrons from photolysis of water.

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Explanation: The main events in the light-dependent reactions occurs in two photosystems, photosystem I and photo system II.

✓ **Photo system II:**

- **Electrons (e-)** in chlorophyll molecules are excited by the energy in photons of light – they become more energetic. Because of the extra energy they escape from the chlorophyll and pass to the electron acceptor(the primary electron acceptor).
- The reaction center molecule is p 680.
- The energy generated in the chloroplast splits water in to hydrogen and oxygen. This is called **photolysis**.
- The electrons released in **photolysis** replaces the electrons lost from the chlorophyll.
- The electrons pass from the primary electron acceptors to electron transport chain.
- The electrons lose energy as they pass from one electron carrier to the next.
- The energy of electrons lose, powers the hydrogen pump which moves protons from the stroma of the chloroplast to the space inside the thylakoid.This leads to an accumulation of proton inside the thylakoid, which drives the chemiosmotic synthesis of ATP.
- The electrons released at the end of electron transport chain of photo system II are used to replace the excited electrons in the chlorophyll molecules in photo system I
- There is no reduction of NADP in photo system II NADP is reduced in photo system I.

∴ ***Answer C***

75. On which one of the following principles' are most of the anti – HIV drugs currently in use working?

A . Inhibition of enzyme action.
B . Degradation of viral RNA.
C. Digesting of viral particles.
D .Phagocytosis of the virus.

Explanation: There are a number of drugs – called **anti-retro viral drugs** that can be effective in slowing down the progressive to AIDS .These drugs work by blocking the reproduction of the HIV in the CD_4 lymphocytes. There are several drugs that can act in different ways at different stages of the cycle of reproduction. The drugs action different stages of the HIV life cycle by inhibiting the enzymes action at different stages of the HIV reproduction. For example some drugs inhibits the enzyme reverse transcriptase that converts viral RNA to DNA.

∴ ***Answer A***

76.Among the following mating, where the ABO blood genotypes of the partners are shown, identify the mating in which all the children will have the same blood type.

A .AO× BO
B .AA× OO
C.AB × BO
D .BB × AO

Explanation:

❖ Parental blood type : A and B
Parental gene type : $I^AI^O \times I^BI^O$
Gametes ..... $I^A, I^O \times I^B, I^O$
Fertilization

	I^A	I^O
I^B	I^AI^B	I^BI^O
I^O	I^AI^O	I^OI^O

All the four possible blood types: A, AB, B, and O

❖ Parental blood type : A and O.
Parental genotype ..... $I^AI^A \times I^OI^O$
Gametes .....I^A and I^O
Fertilization

	I^A	I^A
I^O	I^AI^O	I^AI^O
I^O	I^AI^O	I^AI^O

All the **children** have the same blood type A.

❖ Parental blood type AB and B
Parental genotype ..... I^AI^B and I^BI^O.
Gametes ..... I^A, I^B and I^B, I^O.
Fertilization

	I^A	I^B
I^B	I^AI^B	I^BI^B
I^O	I^AI^O	I^BI^O

25% the children have AB blood type
25 % of the children have A blood type.
50 % of the children have B blood type.

❖ Parental blood type ..... B and A.
Parental genotype ..... $I^BI^B \times I^AI^O$.
Gametes..... $I^B \times I^A, I^O$.
Fertilization

	I^A	I^O
I^B	I^AI^B	I^BI^O

50% of the children have AB blood type ,the other 50% of children have B blood type.

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∴ ***Answer B***

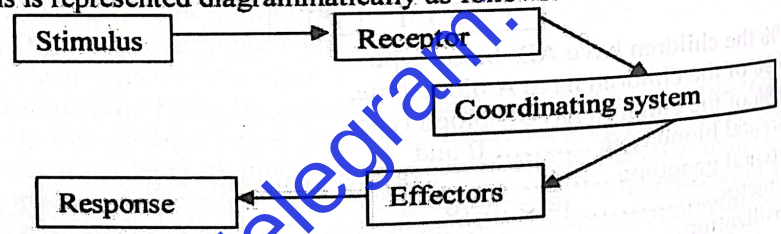
77. If a new mutant allele arises in a certain population, which of the following factors determines if the allele is going to be adaptive or non-adaptive.
A .The environment in which the population lives.
B .The rate at which the gene mutates.
C .The population in which the gene is found.
D .The use and disuse of the gene by the population.
Explanation: **Mutations** are important in introducing variation in to populations. **Mutations** are the ultimate source of variation. The adaptiveness or the non- daptiveness of the new variation introduced by the new mutant allele is determined by the environment in which the population lives.

∴ ***Answer A***

78. Which of the following is the correct route that connects a stimulus and a response?
A Receptor ⟶ coordinator ⟶ effector
B .Coordinator⟶ receptor⟶ effector.
C .Receptor ⟶ effector ⟶ coordinator
D .Effectors ⟶ receptor ⟶ co - ordinator
Explanation: The co - ordinated response of an organism to an internal or external stimulus is a behavior.
For organism to show a co-ordinated response (behavior), then any behavior must have three components:

- **A receptor** of some kind to detect a **stimulus.**
- An **effectors** of some kind to produce the **response** and
- Some kind of linking system or coordinating system that is influenced by the receptor and influence the effectors.

This is represented diagrammatically as follows.



∴ ***Answer A***

79. In which of its contents RNA differs from DNA ?
A. Deoxyribose and guanine.
B .Ribose and uracil
C. Ribose and thymine
D. Phosphate and adenine.

Explanation:

DNA	RNA
Double stranded	Single stranded
Deoxyribose sugar	Ribose sugar
Base thymine instead of uracil	Base uracil instead of thymine.

∴ ***Answer B***

80. Which of the following cell types can be rich in lysosomes?
A .Red blood cells
B .Nerve cells.
C. Phagocytic cells.
D. Muscle cells.
Explanation: **Phagocytic cells** are cells that are capable of engulfing particles or whole cell from their environment by a process termed as phogosytosis.
✓ **Lysosomes** contain digestive enzymes that break down cellular waste and debris. Lysosomes are particularly abundant in phagocytes white blood cells.

∴ ***Answer C***

81. What is the importance of chemiosmosis in photosynthesis and cellular respiration?
A .Splitting of water molecules.
B .Operating the proton pump.
C .Combining hydrogen and carbon.
D .Synthesized ATP.
Explanation: In photosynthesis of electron transport chain the protons (hydrogen ions) are pumped from the stroma of the chloroplast to the space inside the thylakoid. This leads to accumulation of protons inside the thylakoid creating a higher concentration than the stroma. This proton gradient results in the protons diffusion through the ATP syntheses molecule (down the concentration gradient)making the syntheses rotor "spin" and produce ATP from ADP and P1.The
✓ diffusion of hydrogen ions through ATP syntheses is chemiosmosis.
The same thing happens in cellular respiration . Protons (hydrogen ions)accumulated in the intermmbrane space by hydrogen pumps diffuse in to the matrix of the mitochondria (down the concentration gradient) through ATP syntheses molecules resulting production of ATP from ADP and P. This is called **chemiosmosis.**

∴ ***Answer D***

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82. If the age pyramid of a certain country is narrowing at the base , what does this tell about the trend of the population size of the country? The population size is:

A declining
B . growing fast
C. increasing slowly
D. stabilizing.

Explanation: Age pyramid narrowing at the base is a population pyramid that reflects the demographic transition of developed countries. Such population pyramid demonstrates, very low birth rate, low death rate and slow increase in the population size. It also shows an increase in life expectancy and the presence of more aged people than the young.

∴ ***Answer C***

83. Suppose the amino acids coding region in a m RNA is 1200 nucleotides long, how long is the protein in terms of amino acid number?

A .1200 amino acids
B .600 amino acids
C.400 Amino acids
D. 300 Amino acids

Explanation: The m RNA carries code for protein syntheses in its sequence of nucleotides or bases. It is a group of three nucleotides that code for a single amino acids in the protein mRNA molecules which is 1200 nucleotide long carries 400 codes $\frac{1200}{3} = 400$) for 400 amino acids. The protein synthesized by this particular m RNA is 400 amino acids long.

∴ ***Answer C***

84. If a new anti – HIV drugs is to be developed to prevent the virus from entering the host cell, which one of the following processes should the drug target?

A .Reverse transcription
B .Binding of Gp 120 and CD4
C .Integration of viral DNA into host DNA.
D .Assembly of viral parts into a whole virus.

Explanation: Any drug that prevents the entry of HIV in to the T- helper cells (CD_4 cells) is a drug capable of preventing the binding of gp120 (glycoprotein 120) with the CD4 protein receptor on the surface of T- helper cells .HIV enters the T – helper cells only when gp120 of the virus binds with CD4 protein receptor of the host cell.

∴ ***Answer B***

85. Which one of the following factors is **NOT** important for evolutionary change of a population?

A .Over reproduction.
B .Insufficiency of natural resources.
C .Existence of heritable variation.
D .Survival of all that are born.

Explanation:

- **Over reproduction**: all species tend to produce more offspring that can possibly survive.
- **Insufficiency of natural resources:** There will be a "struggle for existence " between members of a species- because they over reproduce, and resources are limited.
- **Existence of heritable variation**: Some members of a species will be better adapted than others to their environment(because there is variation in the offspring).
- **Survival of all that are born:** This does not happen in nature according to Darwin theory of evolution. Because of presence of limited resources the offspring of a species compete for the limited resources. Those which can obtain the resources can survive but the others perish. Besides presence of limited resources there are variations among these offspring and some of the off spring with adaptive variation survives and the others with non- adaptive variation do not survive.

∴ ***Answer D***

86. One of the following would be harder to see under the ordinary light microscope that is more likely to be available in school laboratories.

A .Nucleus
B .A bacterium
C.A mitochondrion
D.A ribosome.

Explanation: It is possible to observe the nucleus of plant or animals cells using special stains under ordinary light microscope.

✓ **A mitochondrion** cannot be seen easily with a light microscope. It may appears as a dot. But it is impossible to see the ribosome's using ordinary light microscope because they are the smallest organelles of the cell.

∴ ***Answer D***

87. In some human liver cells there are 92 chromosomes per cell. What is the ploidy level of such cells?

A .Haploid
B .Diploid
C . Tetraploid
D. Hexaploid

Explanation:

- **Haploid cells** are cells with only one set of chromosome (n number) such as egg and sperm cells in human which contain 23 chromosomes.
- **Diploid cells** are cells having two sets of chromosomes , one set from the male parent and the other set from the female parent.
- ✓ All human body cells contain 46 chromosomes (2n number).
- **Tetraploid** is a cell having four sets of chromosomes (4n number)like some liver cells conting 92 chromosomes.

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❖ **Hexaploid cells** are cells having six sets of chromosomes (6n number).

∴ ***Answer C***

88. Which of the following groups of substances are all inorganic?
A .Water, Sugar, Calcium carbonate.
B .Sugar , fatty acid , amino acids.
C .Water, calcium carbonate, carbon dioxide.
D .Carbon dioxide , amino acid , fatty acid.
Explanation: **Organic molecules** always contain both carbon and hydrogen. In organic molecules may contain one or the other (or neither), but not both.

❖ H_2O , $C_6H_{12}O_6$, $CaCO_3$

❖ $C_6H_{12}O_6$; $R-COOH$, $H_3N^+ - \underset{\displaystyle R}{\overset{\displaystyle H}{\underset{|}{\overset{|}{C}}}} - COOH$

❖ H_2O, $CaCO_3$, CO_2

❖ CO_2, $H_3N^+ ——$ $\underset{\displaystyle R}{\overset{\displaystyle H}{\underset{|}{\overset{|}{C}}}} ——$ $COOH$ - , R - COOH

∴ ***Answer C***

89 .In which of the following compounds are both members of the pair are polymers of carbohydrates that naturally occur in plants only?
A .Starch and chitin
B .Starch and cellulose
C. Glycogen and cellulose
D. Glycogen and chitin.
Explanation: **Starch** is a polymer of ∝- glucose. It is a major fuel store in plants , but absent from animals.

- **Chitin** is closely related to cellulose in structure and function being a structural polysaccharide. It occurs in fungi and in some animal group as component of exoskeleton in insects.
- **Cellulose** is a polymer of carbohydrate β - glucose. It has a structural role in plant cell walls.
- **Glycogen** is the animal equivalent of starch being a storage polysaccharide made from glucose.

∴ ***Answer B***

90.Which of the following is generally expected to give better health benefits when present in human diet?
A .Animal fat
B .Saturated fatty acid
C. Monounsaturated fatty acid
D .Polyunsaturated fatty acid
Explanation:

❖ **Animal fats :** contain a high level of cholesterol which might laid down in the lining of arteries resulting arthrosclerosis'.
❖ **Saturated fatty acids (saturated fats):**causes a rise in blood cholesterol which may result again arthrosclerosis' .

❖ **Mono unsaturated fatty acid** is more or less similar to saturated fatty acids resulting the same health problem.
❖ **Poly unsaturated fatty acids** found in unsaturated fats, are thought to help reduce cholesterol levels in the blood and therefore beneficial to health.

∴ ***Answer D***

91. Where does the light dependent reaction of photosynthesis occur in the chloroplast?
A .In the thylakoid membrane.
B .In the fluid of the stroma.
C .In all parts of the chloroplast.
D .In the stomatal opening.
Explanation: In the light dependent reaction as electrons are transferred from one electron carrier to another, they lose energy .The energy they lose powers the pump which moves protons (hydrogen ions) from the thylakoid membrane of the chloroplast to the space inside the thylakoid.
These processes as indicated in the above paragraph shows that the light dependent reaction of photosynthesis occurs in the thylakoid

∴ ***Answer A***

92. For which of the following is the sugar produced by photosynthesis **NOT** used?
A .To produce biomass.
B To make new DNA.
C. To produce ATP in respiration
D. To produce enzymes.
Explanation:

❖ The sugar produced by photosynthesis is used to produce the biomass of a given area by process of assimilation occurring in individual organisms of the area.
❖ To make new DNA – DNA contains sugar (pentose sugar) in its nucleotide which are the building blocks of DNA.
❖ To make ATP in respiration. In respiration ATP is made from ADP and P_i ,ADP (Adenosine diphosphate) contains a pentose sugar.
❖ Sugars are not used to make enzymes which are proteins. Enzymes or proteins are synthesized from amino acids.

∴ ***Answer D***

93. In which process is ATP generated during short distance high speed running?
A .Aerobic respiration
B .Mitochondrial energy transformation
C .Anaerobic respiration

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D .The Krebs cycle.

Explanation: Many animals carry on anaerobic respiration at certain times. For example, athletes use an enormous amount of energy during a race. They cannot obtain enough oxygen to respire by normal aerobic respiration, and they do not have any special stores of energy. In such occasions the muscle cells **"switch"** to anaerobic respiration. At this time glycogen which is stored in muscles is converted to glucose. Glucose is then broken down without oxygen to produce large quantities of lactic acid and very little energy.

$$\text{Glucose} \xrightarrow{\text{E nzymes}} 2\text{lactic acid} + 2\ \text{ATP}$$

∴ ***Answer C***

94. Which one of the following choices shows the end products of a mitotic cell division?

A .Gametes and ordinary cells.
B .Four genetically different cells
C .Two genetically different cells.
D .Two genetically identical cells.

Explanation: **Mitosis** is the process by which a cell nucleus divides to produce two daughter nuclei containing identical sets of chromosomes to the parent cell and to each other.

Division of the whole cell to form two daughter usually follows it immediately. The two daughter cells identical to one another and to their mother cell (original cell).This is known as **cell division**.

∴ ***Answer D***

95. What exactly happens during a substitution mutation?

A .One base is replaced by a different base.
B .An extra base is added during DNA replication.
C .A base is missed out during DNA replication.
D .Changes occur in the arrangement or structure of a chromosomes.

Explanation: **A mutation** is any spontaneous change in the genetic material of an organism. The change involving a single base are called **point mutation** , and it is these that we are mainly concerned with.

✓ There are several types of point mutation , in which one of the bases in the DNA sequence of the gene is altered, usually by being copied wrongly while the DNA replicates. The different point (gene) mutations are.

- Substitution
- addition
- deletions.

Substitution mutations occur quite randomly .In substitution mutations one base is replaced by a different base. A substitution of just one base

in the sixth triplet of a gene coding for one of the four polypeptides in the hemoglobin molecules alters the triplet from GAG to GTG .This results in the amino acid valine replacing glutamate in the polypeptide chain. The different hemoglobin molecules formed results in the condition known **as sickle cell Anemia.**

A point mutation in which an extra base is added during DNA replication is **addition mutation.**

- A point mutation in which a base is missed out during DNA replication is a deletion mutation.
- Changes that occur in the arrangement or structure of a chromosomes results in chromosomal mutations

∴ ***Answer A***

96. During the course of evolution, which of the following events came before all the others?

A .Origin of dinosaurs and other reptiles.
B .Availability of free oxygen in the atmosphere.
C .Origin of the oldest eukaryotic organisms.
D .Origin of the first multicellular animals and plants.

Explanation:

- Dinosaurs originated between 200 to 250 million years ago.
- It was by 2.4 billion years ago the concentration of oxygen began to rise and the rate of increase accelerated from 2.1 billion years ago.
- Eukaryotes become numerous 1.9 to 2.1 billion years ago.
- First multi cellular animals evolved perhaps 600 to 700 million years ago.

∴ ***Answer B***

97. What kind of natural selection is at work in a situation where individuals at both extreme ends of the range get better survival advantages than those around the middle?

A .Stabilizing selection.
B .Directional selection.
C. Disruptive selection.
D. Selection that removes the extremes.

Explanation:

- **Stabilizing selection**: In a stable environment individuals at both ends of the range of values for a feature are at least well adapted. Selection

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often operates against both these extremes to reduce the variability in the population and to make the population more uniformly adapted.

- **Directional selection:** A feature may show a range of values. Individuals at one extreme could have a disadvantage where those at the other extreme have an advantage. Over time , selection operates against the disadvantaged extreme and in favor of the other extreme. The mean and range of values shift towards the favored extreme.
- **Disruptive selection:** is in effect, the converse of stabilizing selection. In this instance, individuals at both extremes of a range have some advantage over those displaying the mean value. As a result, the frequency of those individuals at the extremes of the range will increase over time and those in the middle of the range will decrease over time.
- **Selection that removes the extremes** is stabilizing selection.

∴ ***Answer C***

98. In which geologic period does the fossil record show more diverse and relatively higher forms of organisms?

A .Devonian C. Jurassic
B .Cretaceous D. Permian.

Explanation:

- **Devonian:** a geological period in which the first amphibians and jawed fishes diversity occurred.
- **Cretaceous:** a geological period in which extinction of dinosaurs, first primates and first flowering plants occurred.
- **Jurassic:** a geological period in which the first birds and dinosaurs diversity occurred .
- **Permian:** a geological period in which major extinctions and reptiles diversity occurred.

∴ ***Answer B***

99. To what kind of animal behavior can the spinning of a web by a spider be classified?

A .Learned behavior C. Instinctive behavior
B .Experiential behavior D. Accidental behavior.

Explanation: **Instinctive behaviors' are pre:** programmed patterns of behavior. They are not just single action in response to a simple change in the environment. **Instinctive behaviors'** often involve a complex sequence of action. A good example to illustrated this point is the

spinning of a web by spiders. It involves many complex actions yet the spider does not have to learn how to do it. It spines a perfect web the very first time- as well as every other time after wards.

∴ ***Answer C***

100. Which of the following is a recently developed active area of research in biology today?

A .Taxonomic study C. Ecological research
B .Study about the cell theory. D. Stem cell research.

Explanation:

- **Taxonomic study** the science of classifying animals and plants. Modern taxonomic classification based on the natural concepts of the Swedish botanist carolus Linnaes, has progressed steadily since 18th century.
- **Cell theory** the theory that all organisms are composed of cell products and that growth and development results from division and differentiation of cells. This idea resulted from numerous investigations that started at the beginning of the 19th century and it was finally give form by Schleiden and Schwann in 1839.
- **Ecology** is the study of the relationships of living organisms with each other and their non-living or physical surroundings.
- **Stem cells :**Undifferentiated cells in embryo or adult which can undergo unlimited division and can give rise to one or several different cell types. Based on this fact stem cell research is one of the 21st century scientific research which may lead to the production clone of various organs from stem cells.

∴ ***Answer D***

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1. Which one of the following molecules in living things is **NOT** an organic molecules ?

A .Sugar
B .Nucleotide
C. Amino acid
D. Water

Explanation: **Organic molecules** always contain both carbon and hydrogen. **Inorganic molecules** may contain one or the other (or neither),but not both.

- Sugar ($C_6H_2O_6$)
- **Nucleotide** : the monomer of the polymer DNA.
- Amino acid ($H_3N^+ - \underset{\underset{R}{|}}{\overset{\overset{H}{|}}{C}} - COOH$)
- Water(H_2O)

∴ Answer D

2. What is the general term for the part of the protoplasm that lies outside the nucleus?

A . Cytosol
B . Cytoplasm
C. Central vacuole
D. Plasma membrane.

Explanation:

- **Cytosol** is the cytoplasm other than the various membrane – bounded organelles.
- **Cytoplasm** is the living contents of a cell, excluding the nucleus and large vacuoles , in which many metabolic activities occur.
- **Central vacuole:** is found in plant cells. The vacuole in a plant cell is a fluid –filled sac that stores a range of solutes.
- The **membrane** that surrounds and encloses a cell is sometimes called the cell surface membrane, but most biologists now refer to its as **plasma membrane**. It is selectively (partially) permeable membrane.

∴ Answer B

3. Which of the followings do viruses share with cells?

A .Cell wall
B .Cytoplasm
C. Nucleic acid.
D. protoplasm

Explanation: The various particle or virion , is , in its simplest form , just **nucleic acid** enclosed in a protein shell (protein coat). Viruses cannot independently carry out any of the processes common to all living organisms. They can only reproduce inside other cells. So they are all parasites. The basic virus is not even a cell- it has no nucleus and cytoplasm but it does have genetic material (nucleic acid) surrounded by a protein coat.

∴ Answer C

4. Which of the following structures is **NOT** present in animal cells?

A .Cell wall
B .Nucleus
C. Protoplasm.
D. Nucleic acid

Explanation:

- **Cell wall:** non- living rigid structure surrounding the plasma membrane of algae, plant ,fungal and bacterial cells but not animal cells.
- **Nucleus:** An organelle of plant and animal cells (eukaryotes) containing the genetic information (DNA) and hence controlling cell's activities.
- **Protoplasm:** the term once used to describe the ground substance – the living material of cells. This material would include the complex colloidal organization of substances making up a cell nucleus, cytoplasm, plastids and mitochondrion.
- **Nucleic acid:** Organic acids whose molecules consists of chain alternating sugar and phosphate units, with nitrogenous bases attached to the sugar units, they occur in the cells of all organisms.
- DNA and RNA are nucleic acid.

∴ Answer A

5. What is the branch of biology that studies about the origin and gradual changes of living things?

A .Microbiology
B .Evolution
C. Mutation.
D. Reproduction

Explanation:

- **Microbiology** is the study of microscopic organisms (micro-organisms),including their interactions with other organisms and with their environment.
- **Evolution** in biology, desent with modification, the process whereby all living things on earth have diverged, by direct desendant from a single origin of life that occurred more than 3,000 million years ago.
- **A mutation** is any spontaneous change in the genetic material of an organism.
- **Reproduction** is a process whereby living plant or animal cells or organisms produce offspring. **Reproduction** is one of the essential functions of living organisms as necessary for the preservation of the species as food getting is for the preservation of the individual.

∴ Answer B

6. Which one of the following cellular forms did Robert Hooke observe under his crude microscope?

A .Bacteria
B .Protozoa
C. Yeast
D. Empty cell wall.

Explanation: **Robert Hooke** using crude microscope was able to examine the section of a cork (dead tissue).During his observation he

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noted that the cork was made up of numerous box-like structures(empty cell walls),and he called these structures cells for the first time.

∴ ***Answer D***

7.Behavior that is repeated on a daily basis is referred to as:

A .Lunar C. Circannual

B .Circadian D. Seasonal.

Explanation: Both plants and animals show yearly, monthly ,daily and other cyclical changes that are genetically programmed.

✓Daily rhythms are called **circadian rhythms.** Circadian clocks have two main featness:

✓They will persist within a period of about 24 hours in the absence of environmental cues.

✓They can synchronise to a 24 hour cues such as light – dark cycle; this is called entrainment.

✓Changes in the light – dark ratio can control reproductive behavior on an annual basis. Such rhythms are called **circannual(yearly)** rhythms. Seasonal rhythms are circannual rhythms.

∴ ***Answer B***

8.Which molecule in the cell is the constituent of the gene?

A .Nucleic acid. C. Lipid

B .Protein D. Carbohydrate.

Explanation:

❖ **Nucleic acids** :are polymers of monomers called **nucleotides**. There are two types of nucleic acid, DNA and RNA .DNA is genetic material of all organisms .A gene is a section of DNA that determines a particular trait of organism.

❖ **Protein** :One of large number of substances that are important in the structures and function of all living organisms .Proteins are polypeptides. i.e they are made up of amino acids molecules joined together by peptide link. All enzymes are proteins.

❖ **Lipids:** are diverse group of fatty substances found in all living organisms. Lipids are distinguished from other classes of organic compounds in that they do not dissolve in water but are soluble in alcohol, ether, or other organic solvents.

❖ **Carbohydrates:** A class of compounds occurring widely in nature and having the general formula type $C_X(H_2O)Y$.Carbohydrates are both stores of energy and structural elements in living systems. They are composed of carbon, hydrogen and oxygen.

∴ ***Answer A***

9. While conducting research, which of the following should a biologist do first?

A .Generate hypothesis

B .Conduct experiments.

C .Communicate the results.

D .Identify a research question/problem.

Explanation: Scientific experiments are carried out based on a specific problem that has to be identified before proceeding to the next steps. Observations are made and facts gathered leading to the definition and elaboration of the problem or question more precisely.

- After gather information about a particular problem a hypothesis can be generated. A hypothesis is a suggested explanation of certain observed phenomena.
- A hypothesis formed must be tested using experiments.
- The experimental results obtained is communicated. Communication is an important component of scientific method. It is a way of reaching the scientific community. The proper channels of communication for scientists and researchers are scientific journals.

∴ ***Answer D***

10.In the case of enzymatic chemical reactions, what do you call the substance that is acted upon by an enzyme?

A .Polypeptide C. Vitamin

B .Coenzyme D. Substrate

Explanation:

❖ **Polypeptide:** A compound that contain many amino acids linked to gather by peptide bonds.

❖ **Coenzyme :** A non-protein group without which certain enzymes are inactive or incomplete.

❖ **Vitamins:** Organic chemical compounds that are essential in small quantities for metabolism. Vitamins have no energy value; most of them seen to act as catalysts for essential chemical changes in the body.

❖ **Substrate :** The substance up on which an enzyme act.

∴ ***Answer D***

11.On which of the following organelles of the eukaryotic cell does protein synthesis take place?

A .The nucleus C. The chloroplast

B .The ribosome D. The mitochondrion.

Explanation:

❖ **Nucleus :**An organelle of plant and animal cells(eukaryotes) containing the genetic information (DNA) and hence controlling cell activities.

❖ **The ribosome :** A small organelle found in all cells on which protein is synthesized. They are known as protein factories , because they are the sites of protein synthesis.

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❖ **The Chloroplast :** A photosynthetic plastid containing chlorophyll and other photosynthetic pigments. It is found in all photosynthetic plant cells. It is the site of carbohydrate synthesis.

❖ **The mitochondrion:** An organelle of all plant and animal cells chiefly associated with aerobic respiration.

∴ ***Answer B***

12. Which of the following is a distinguishing feature of amino acids that is **NOT** found in carbohydrates?

A .Carbon
B .Oxygen
C. Hydrogen
D. Nitrogen.

Explanation: **Carbohydrates** are made up of carbon, hydrogen and oxygen.

✓ **Protein** are made up of **carbon ,hydrogen, oxygen and nitrogen** (which is absent in carbohydrates).

∴ ***Answer D***

13. Choose the organisms that belongs to the eukaryotes?

A .Bacteria
B .Blue- green algae.
C. Amoeba
D. Virus

Explanation:

❖ **Bacteria and blue – green algae** (also known as cynobacteria) are prokaryotes.

❖ **Amoeba :** are eukaryotic cells that belong to the group protozoa.

❖ **Viruses** are not cellular (they are not cell at all).

∴ ***Answer C***

14. An experimental animal stopped responding to a stimulus that has been repeated so many times. What type of learned behavior is this?

A .Latent learning
B .Habituation
C. Imprinting
D. Operant conditioning

Explanation:

❖ **Latent learning** is a Learning that is not apparent as it takes place. Latent learning happens when the brain acquires knowledge at a certain time, without reinforcement, but does not use it until later, at a time when the knowledge is needed.

❖ **Habituation** is a process which results in decreased response to a stimulus after repeated exposure to that a stimulus.

❖ **Imprinting** is a kind of instinctive behavior in which the fixed action pattern is for newly born /hatched organism to imprint on (or become attached to)the first thing they see that has certain general features (those of an adult of its species)

❖ **Operant conditioning** is a classical conditioning involves modifying an innate response by pairing it with a previously neutral stimulus. Operant conditioning can modify more complex, voluntary behavior by the

animals /person learning to associate the behavior with certain specific consequences .It is learning by trial and error

∴ ***Answer B***

15. How many net ATP molecules are generated through anaerobic respiration, when a single glucose molecule is changed to pyruvate in the human body?

A .Two
B .Three
C. Four
D. Six.

Explanation: **Glycolysis** is anaerobic breakdown of carbohydrates, specifically glucose ,to pyruvic acid. At the end of glycolysis, there is a net gain of two ATP molecules per molecules of glucose. Glycolysis in general produces four ATP molecules but two ATP molecules are consumed in the process of these chemical reactions.

∴ ***Answer A***

16. Which of the following statements is true about matter and energy in the ecosystem?

A .Both matter and energy are recycled.
B .Matter is not recycled ; energy is recycled.
C .Matter is recycled; energy is not recycled.
D .Energy is recycled more than matter does.

Explanation: **Recycling in nature** involves the continuous exchange of material between the living and non-living components of the ecosystem. All of the substances that occur with organisms, including water, carbon, nitrogen ,oxygen and others (sulphur, phosphorus etc) cycle through ecosystems. Without recycling of these elements, life in the ecosystem could not have existed continually.

✓ **Energy** ,like that of materials pass from one trophic level to the next in an ecosystem. The flow (transfer) o f energy unlike that of materials is one directional, i.e energy cannot be recycled ,while material do. In other words ,we do not say energy cycle.

∴ ***Answer C***

17. Which of the following groups of bacteria consists of rod shaped cells?

A .Gram-positive
B .Bacilli
C. Cocci
D .Spirilla.

Explanation:

❖ **Gram positive bacteria:** are stained purple by gram's stain. The bacterial cell wall is largely composed substances known as **peptidoglycan.** Gram positive bacterial have much more peptidoglycan in their cell wall which enable them to retain the crystal violet stain that staines them purple.

❖ **Bacilli- (Singular – bacillus)** are rod shaped bacteria.

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- **Cocci – (Singular coccus)** are Spherical or round shaped bacteria – oooo
- **Spirilla (singular spirillum)** are spiral shaped bacteria.

∴ ***Answer B***

18. Which of the following parts of plant roots is harbouring nitrogen fixing bacteria?
A .Root nodule
B .Root hair
C. Root tip
D. Root cap.

Explanation: **Nitrogen fixing bacteria** are bacteria that convert the nitrogen gas (N_2) into ammonium ion in the nitrogen cycle of an ecosystem. Nitrogen fixing bacteria are found in the root nodules (swellings) of legumes.

∴ ***Answer A***

19. How many carbon and oxygen atoms are there in a molecule of maltose?
A .6Carbon and 6 oxygen.
B .12 Carbon and 11 Oxygen
C.11Carbon and 12 oxygen.
D. 24 Carbon and 12 Oxygen.

Explanation: **Maltose** is a disaccharide made from two glucose units which contains 6 carbon,6 hydrogen and 6 oxygen ($C_6H_{12}O_6$) . When maltose is formed from two glucose molecules by condensation reaction 2 atoms of hydrogen and one atom of oxygen are removed as water. Therefore, a molecule of maltose contain 12 carbon atoms, 22 hydrogen atoms and 11 oxygen atoms.

∴ ***Answer B***

20. Which of the following is true about protozoa?
A .They are a group of bacteria.
B .They are one—celled plants.
C .They represent both plants and animals.
D .They are unicellular animals.

Explanation: **Protozoa** –in zoological classification ,a phylum of unicellular heterotrophic, generally non photosynthetic, aquatic eukaryotes lacking cell wall.

∴ ***Answer D***

21. Which of the following carbohydrates has structural function?
A .Cellulose
B .Glycogen
C. Starch
D. Sucrose.

Explanation:

- **Cellulose** is a polymer of β - glucose .Unlike starch and glycogen it has a structural role. About 50 % the carbon found in plant is cellulose .Cellulose is one of the major components of plant cell walls.

- **Glycogen** is the animal equivalent of starch, being a storage polysaccharide made from glucose .In vertebrates glycogen is stored chiefly in the liver and muscles.
- **Starch** is a polymer of α - glucose. It is a major fuel store in plants but is absent from animals where the equivalent is glycogen.
- **Sucrose or cane sugar** is the most abundant disaccharide in nature. It is commonly found in plants.

∴ ***Answer A***

22. Which of the following cellular structures is possessed by all cells?
A .Cell membrane
B .Nucleus
C. Cell wall
D. Golgi apparatus

- **Cell membrane** is also known as **plasma membrane**. The cell, membrane a living part of a cell , is a thin covering found in all plant cells, animal cells and in all other eukaryotic cells and prokaryotic cells.
- **Nucleus** is an organelle of all eukaryotic cells bounded by nuclear membrane. It contains a dense and round body called **nucleolus** and the genetic material – DNA .It is found only in eukaryotic cells.
- **A cell wall** is a structure that forms a semi-rigid covering around plant cells. Plant cells possess a very conspicuous cell wall and while animals do not.
- **The Golgi – apparatus** consists of stacks of plate like membranes filled with cellular products. Its function is to serve in collecting, concentrating and packaging cellular products in the form ready to transport. They are found in both plant and animals cells (all eukaryotic) but not in prokaryotic.

∴ ***Answer A***

23. At which stage is most of the ATP generated in aerobic respiration?
A . Glycolysis
B .Link reaction.
C. Krebs cycle
D. Electron transport.

- **Glycolysis** is the anaerobic phase of the aerobic respiration generate only 2 Net ATP molecule.
- **Link reaction** that joins glycolysis to krebs cycle does not generate ATP.
- **Kerbs cycle the cyclic** stage of aerobic respiration generates only 2 ATP molecules
- **Electron transport chain** , the last stage of aerobic respiration generates 30 ATP molecules.

∴ ***Answer D***

24. Which of the following groups of animals have males with ZZ and females with ZW sex chromosome constitution?
A .Grasshoppers
B .Birds
C. Mammals
D. Honey bees.

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Explanation: **Sex determination** is a system by which the sex of an organism is determined. There are a number of systems of sex – determination:

- **Grass hoppers :** the Y chromosomes may be absent and so the male has XO genotype and the female XX genotype.
- **Birds :** is the female who are the heterogametic sex and the male homogametic sex. Females are ZW and males are ZZ.
- **Mammals :**males are heterogametic and females are homogametic – males have the XY genotype and females XX genotype.
- **Honey bees** the females have XX and the males have XO genotype.

∴ ***Answer B***

25. In some crosses of maize, the progeny produces better yield than the parents. What is this phenomenon known as?

A .Dominance
B .Heterozygosity.
C. Hybrid vigor
D. Inbreeding depression.

- **Dominance :**when two alleles of a pair are different , one is dominant (will be expressed in the organisms appearance or physiology) and one is recessive (cannot be expressed unless the individual has two copies of recessive allele)the dominant traits masks the appearance of recessive traits.
- **Heterozygous :**Having two different alleles at a given locus. Usually only one of these , the dominant allele is expressed in the phenotype. On selfing or crossing heterozygote's some recessive may appear giving viable offspring. Selfing heterozygote's halves the hetrozygosity (being heterozygous) and thus out breeding maintains hetrozygosity and produce a more adaptable population.
- **Hybrid vigor (Hetrosis) :** The condition in which the expression of a characteristics is greater in the heterozygous offspring than either of the homozygous parents.

  Hybridvigor occurs when unrelated breeds or line of the some species are cross – bred .In many cases the off spring from these crosses are more productive than the average of their breeds. The same principles can apply in breeding crops. Cross – breeding natural or planned has been important in producing many of the high yielding crop plants we now use such as maize and others.
- **In breeding depression:** Breeding between closely related individuals. The most extreme form of in breeding is self-fertilization ,which occurs in some plants.

∴ ***Answer C***

26. Which of the following is the common characteristics of all enzymes?

A .They act inside cells only.
B .They are proteins.
C. They require cofactors.
D. They operate at any PH.

Explanation :These are the common characteristics of enzymes :

- ✓ They are all proteins.
- ✓ They are biological catalysts that speed up a reaction without being used up so they can be used over and over again.
- ✓ They are specific; they catalyze one reaction only.
- ✓ A small amount of enzyme can bring about a change in a large amount of its substrate.
- ✓ Enzymes are affected by PH and temperature

∴ ***Answer B***

27. During the Krebs cycle, which of the following molecules temporarily stores most of the energy released from food molecule?

A .ADP
B .ATP
C.NADH
D. FADH

Explanation :**NADH:** is the reduced form of NAD .NAD (nicotine amide adenine dinucleotide) – A derivative of nicotine acid that acts as a coenzyme in electron transfer reactions (Example: the electron transport chain) . Its role is to carry hydrogen atoms. The oxidation of one molecule of reduced NAD (NADH) results the synthesis of three ATP.

- ✓ **FAD (Flavin adenisine dinucleotide) :**The oxidation of one molecule of reduced FAD (FADH) results the synthesis of 2ATP .
- ✓ In Krebs cycle, from a molecule of glucose 6NADH and 2FAD reduced are produced.

∴ ***Answer C***

28. Which of the following forms of nitrogen is readily utilizable by green plants?

A .Atmospheric nitrogen.
B .Nitrite.
C. Ammonium ion.
D. Nitrate

Explanation :

In the nitrogen cycle- decomposers decay the extratory products and detritus, releasing ammonium ions($NH4^+$) in to the soil ;this process is often referred to as ammonification. There are some plants that can absorb (NH^+_4)ammonium ions from the soil to use it as a source of nitrogen.

✓ Nitrifying bacteria oxidize the ammonium ions to nitrates (NO_3^-)in the process called **nitrification** ; in this process there is an intermediate product called nitrite (NO_2^-) **Nitrate** is the form of nitrogen compound that most plants absorb from the soil.

∴ ***Answer D***

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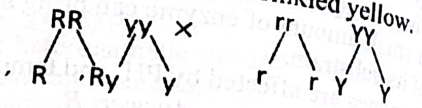
29. In a cross between round green pea of RRyy genotype and wrinkled yellow pea of rrYY genotype, what is the expected genotype of the F_1?
A . RRyy
B . rr YY
C. RRYY
D . RrYy

***Explanation* :Parental phenotype:** round green × wrinkled yellow.
parental genotype



✓ Miosis (gamete formation)
✓ Gametes : Ry × rY

✓ Fertilization : Ry × rY ⇒ Rr Yy
✓ All of the plants of F_1 will have round and yellow seeds(Rr Yy).

∴ ***Answer D***

30. Which of the following techniques is used to separate DNA fragments according to their size on a gel?
A .Southern blotting
B .Electrophoresis
C. Radioactive labeling.
D. Digestion by restriction enzyme

***Explanation* : Electrophoresis:** Gel electro phores is a technique that uses a thick block of gel (jelly – like material) to act as a "molecular sieve." Fragments' of DNA (or protein molecule) are separated by applying an electrical field across the gel. Because DNA fragments are negatively charged ,they move to the positive electrode. The smaller fragments move more quickly than the larger ones and so move further in the sometime.

∴ ***Answer B***

31 .According to Lamarck's theory of evolution , what is the mechanism by which evolving organisms acquire new structures?
A .Mutation
B .Hereditary variation
C .Use- and – disuse of parts of the body.
D .Recombination of ancestral genes.

***Explanation* :**

- **Mutation** is any spontaneous change in the genetic material of an organism.
- **Hereditary variations** are different features or characteristics seen in individuals of member of the same species. Variations that pass from one generation to the next (heritable variations)
- **Lamarck suggest** that by continually using a structure or process ,that structure or process will become enlarged or more developed.

- **Recombination of ancestral genes** : Recombination is the grouping of hereditary factors or genes that regularly occur during meiosis as a result of the independent assortment of chromosomes.

∴ ***Answer C***

32.Which of the following is more true about the male of the honey bee?
A .It has no father.
B .It is sterile
C. It is diploid
D. It regulates the colony

***Explanation* :**The male honey bee also known as drone is produced from unfertilized egg. That is the egg without being fertilized by a male gametes (sperm cell)develops in to a male honey bee. It is haploid , therefore its genotype is XO (no y –chromosomes) .It is fertile. The function of the drone is mating with the queen honey bee.

∴ ***Answer A***

33.Which of the following mechanisms moves digested amino acids and glucose across the plasma membrane of the cells lining the wall of the small intestine?
A .Osmosis
B .Simple diffusion.
C. Facilitated diffusion
D. Active transport.

***Explanation* :**

- **Osmosis** is the process by which water moves across a partially permeable membrane.
- **Simple diffusion** is the movement of molecules or ions from a region of higher concentration to a region of lower concentration down the diffusion gradient.
- **Facilitated diffusion** is essentially the same process as diffusion in that it depends on a concentration gradient to allow particles to cross the membrane. However , it differs in that the particles must be helped to diffuse across the membrane (their diffusion must be facilitated) by a carrier protein or a channel protein within the ion pore.
- **Active transport** is sometimes substance must be moved against a concentration gradient –from a low concentration to higher one .It can only happen if metabolic energy is used to drive the process. Digested substance such as amino acids and glucose moves across the plasma membrane of cells lining the wall of the small intestine by **active transport.**

∴ ***Answer D***

34. If present in the fermentation system, which one of the following would negatively affect alcohol production by yeasts?
A .Water
B .Oxygen
C. Glucose
D. Fermentation enzyme.

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Explanation **:Fermentation** is anaerobic respiration where energy is released anaerobically (without oxygen)from glucose. Yeasts are facultative anaerobs (anaerobs that can respire in the presence or absence of oxygen). Yeast can respire aerobically in bread making but it must respire anaerobically to make alcoholic drinks. In this case presence of oxygen poisons the yeast and stops the process of fermentation.

∴ ***Answer B***

35. Which one of the following is the mechanism by which two genes located on the same chromosome are separated?
A .Independent assortment. C. Linkage
B .Segregation D. Crossing over.

- **Independent assortment :** This is Mendel's law of independent assortment which state that the inheritance of one trait is independent of the inheritance of another. This is where the alleles of one pair sergregate independently of the alleles of another pair controlling different features.
- **Segregation:** The heritable factors (alleles) separate or segregate when gametes (sex cells)are formed ,each gamete therefore, contains only one allele controlling the trait. This is Mendel's law of **segregation.**
- **Linkages :** Some genes are always inherited together with other genes- they exhibit linkage. This happens when the genes in question are on the same chromosomes. A chromosome is a physical unit on it, so when a particular chromosome is passed in to a gamete ,all the genes on that chromosome pass in to the same gamete.
- **Crossing over :**whilst associated in the bivalent (pairing of homologous chromosomes), chromatids from different chromosome undergo crossing over. These chromatids are called non-sister chromatids. In this process the chromatids attached to one another and the exchange equivalent section of DNA (genes), and all the four chromatids in homologogous pair are genetically different .It is a mechanism by which genes located on the same chromosome separate and exchanged with genes located on another chromosome.

∴ ***Answer D***

36. Which one of the following eukaryotic cell organelles was a free living cell before eukaryotic cells evolved?
A .Nucleus C. Chloroplast
B .Ribosome D. Nucleolus.

Explanation :According to **Endosymbiont** theory of the origin of eukaryotic cells, the prokaryotic cell that become membraneous engulfed other smaller cells that were better at respiring organic molecules to release energy. These **engulfed** prokaryotes would evolve

in to the mitochondria of eukaryotic cells. One further stages suggests that some of these cells with their " **primitive mitochondria**" also engulfed, smaller prokaryotic cells. These were prokaryotic cells that could photosynthesis and would in time evolve in to chloroplast. The cells that contained these would be autotrophs and the for runners of plants cells.

The primitive eukaryotic cell that engulfed another prokaryotic cell was a cell containing DNA like the existing nucleus in eukaryotic cells.

∴ ***Answer A***

37. The molecule of which pigment is located at the reaction center of a photo system?
A .Chlorophyll a C. Carotenoid
B .Chlorophyll b D. Accessory pigments

Explanation **:** A photo system consists of a number of pigmented molecules all clustered around one particular chlorophyll molecule called the reaction center molecule. This cluster of pigment molecules is called an antenna complex. The pigment in the antenna complex include **chlorophylla, chlorophyll b** and **cartotenoids.** The reaction centre molecule is always **chlorophyll a.**

∴ ***Answer A***

38. In Ethiopia, which one of the following vertebrate groups has the largest number of genera?
A .Mammals C. Reptiles
B .Birds D. Amphibians.

Explanation **:**

- **Mammals** in Ethiopia have 144 genera.
- **Birds** in Ethiopia have 306 genera.
- **Reptiles** in Ethiopia have 36 genera.
- **Amphibians** in Ethiopia have 19 genera

∴ ***Answer B***

39. In which one of the following ways the Tundra biome differs from desert biome?
A .It is a treeless biome
B .It has little biodiversity
C. It receives little precipitation.
D. It has permanently frozen subsoil

Explanation :

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Biome	Rain fall	Temp	Soil	Plants	Animals
Desert	Almost none	Hot	Poor	Sparse succulent (like cacti)	Sparse, insects arachnids reptiles and birds.
Tundra	Dry	Cold	Perma frost	Lichens and mosses	Migrating animals.

∴ ***Answer D***

40. Which of the following substances is **NOT** necessary for photosynthesis to take place?
A .Chlorophyll
B .Carbon dioxide
C. Oxygen
D. Water

Explanation :
- **Chlorophyll** is a green pigment that traps sunlight energy in the process of photosynthesis.
- **Carbon dioxide** is an in organic raw material used to synthesis carbohydrate.
- **Oxygen** is the byproduct of photosynthesis produced by splitting water molecules in to hydrogen and oxygen in the light dependent phase of photosynthesis.
- **Water** is an inorganic raw material that is splited in to hydrogen and oxygen in the light dependent phase of photosynthesis hydrogen makes the hydrogen of carbohydrates.

∴ ***Answer C***

41. In which of the activities of the honey bee colony are worker bees involved?
A .Laying eggs
B .Serving as queens
C. Reproductive functions
D. Taking care of the larvae.

Explanation :The worker honey bee are involved in the following activities of the honey bee colony:
- cleaning up dirty honey comb.
- Feeding larvae (taking care of larvae)
- Receiving food from incoming foragers (food gatherer worker bees)
- Remove corpses
- Foraging for nectar ,pollen and water

∴ ***Answer D***

42. Where exactly in the cell does the Krebs cycle take place?
A .Mitochondrial matrix
B .Inner mitochondrial membrane.
C .Cytoplasmic fluid.
D .Outer mitochondrial membrane.

Explanation :
- **Mitochondria matrix** is a center of a mitochondrion where krebs cycle takes place.
- **Inner mitochondrial membrane** is a site of electron transport chain.
- **Cytoplasmic fluid** is a site of glycolysis.
- **Glycolysis** is anaerobic respiration in which glucose is broken down to pyruvate.
- **Outer mitochondrial membrane** is a membrane surrounding the mitochondrion on the out side. This is not involved in any process of respiration.

∴ ***Answer A***

43. Which of the following can be taken as case for evidence of evolution from the field of plant & animal breeding?
A .Artificial breeding always creates new species.
B .Artificial breeding only produces new organisms with higher yield.
C . Artificial breeding shows that natural selection can produce variations.
D .Artificial breeding can produce new variations in organisms.

Explanation :Over many generations , selective breeding can bring about significant changes to the organisms involved.
If new varieties can be produced by selective breeding (in which humans choose which individuals will survive to breed) then natural selection (in which environmental process select(which will survive to breed)should also be able to produce new varieties and eventually , new species

∴ ***Answer C***

44. If a radioactive substance that weighs one kilogram has a half – life of 100 years, what would be the percentage of the substance left after 300 years?
A .50 B .30 C.25 D.12.5

Explanation : If a radioactive substance that weighs one kilogram has a half life of 100 years , at the end of the first half life only 50% of the original 1 kilogram will remain. At the end of the second half (200 years) only 50 % of the remaning or 25% of the original will remain.Finally at the end of the third –half life (300 years) only 50% of the remaning or 12.5% of the original will **remain** (25 k.gm)

∴ ***Answer D***

45. Which of the following PH values represents the strongest base?
A .2 B.7 C.13 D.15

Explanation :**PH** is a measure of the acidity or alkalinity of a solution on a scale 0 -14 .**A neutral solution** has a PH of 7. **Acid solutions** have a PH below 7; alkaline solutions have a PH above 7.As the PH

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number decreases from 7 to 0 acidity increase and similarly as PH number increases from 7 to 14 ,alkalinity of the solution increases

∴ ***Answer C***

46. Which of the following is an accurate representation of the relationships between the terms metabolism, anabolism and catabolism?
A .Anabolism = catabolism.
B .Metabolism = Catabolism
C .Anabolism = catabolism + metabolism
D .Metabolism = Catabolism + Anabolism.

Explanation :**Metabolism** is an integrated network of biochemical reactions in living organisms often referring to the biochemical changes occurring in the living organisms or cell as a whole. It includes anabolism and catabolism in metabolic path way.

- **Anabolism :** the constructive biochemical process in living organisms involving the formation of complex molecules from simpler ones and the storage of energy.
  Example : photosynthesis , protein synthesis etc.
- **Catabolism :** is the breaking down of complex molecules by living organisms with release of energy.
  Example :Digestion ,respiration etc.

∴ ***Answer D***

47. Choose the characteristics that viruses share with living things.
A .They are made up of many specialized cells.
B .They contain genetic material.
C .They reproduce by mitosis.
D .They contain chlorophyll.

Explanation :**Viruses** cannot independently carry out any of the process common to all living organisms .They can only reproduce inside other cells. So they are all parasites. The basic virus is not even a cell – it has no nucleus and no cytoplasm but it does have genetic material surrounded by a protein coat.

∴ ***Answer B***

48 .Which group of organic compounds includes the enzymes?
A .Proteins
B .Lipids
C. Starches
D. Carbohydrates

Explanation :Refer to question number 8.

∴ ***Answer A***

49. Into which of the following organic compounds can lipase, maltase and sucrase be grouped?
A .Hormones
B .Carbohydrates
C. Nucleic acids
D. Enzymes.

Explanation :**Lipase** is a hydrolytic enzyme that breaks down fats in to three molecules of fatty acids and one molecule of glycerd.

- **Maltase** is a hydrolytic enzyme that breaks maltose in to two molecules of glucose.
- **Sucrase** is also a hydrolytic enzyme that breaks sucrose in to a molecule of glucose and fructose.

∴ ***Answer D***

50.What is the ultimate source of energy for all organisms living on the earth?
A .Water
B .Oxygen
C. Nitrogen
D. The sun

Explanation :In the ecosystem , the movement of energy is not cyclic but unidirectional. It flows from the sun(the primary source) to the producers, which can transform solar energy in to chemical energy . This is why the sun is said to be the ultimate source of energy for all living organisms on the planet

∴ ***Answer D***

51. Which of the following is true about the substance represented by the chemical formula $C_2H_5NO_2$?
A .Simple sugar
B .Inorganic substance
C. Fatty acid
D. Amino acid

Explanation :

- **Simple sugars** are monosaccharide's. Their general formula is $(CH_2O)n$.They are classified according to number of carbon atoms as trioses (3C) ,tetroses (4C) , pentoses (5C), hexoses (6C) and heptoses (7C) .Of these;pentoses and hexoses (glucose) are the most common.
- **Inorganic substances** are materials or molecules that may contain carbon or hydrogen but not both.
- **Fatty acids** contain the acid group - COOH (the carboxyl group)and are so named because of the larger molecules in the series occur in fats.
- **Amino acids** are organic acids that contain the COOH (carboxyl group).In addition , they each have amino group (NH_2).

∴ ***Answer D***

52. Which of the following is **NOT** a term used to describe organisms that have had foreign genes added to them ?
A .Genetically modified organisms
B .Transgenic organisms.
C .Genetically engineered organisms.
D .Pathogenic organisms.

Explanation :Genetically modified organisms, transgenic organisms and genetically engineered organisms are the biological terms or phrases used to describe the same process of a modern biotechnology which is

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the transfer of genes from one organism in to the DNA of another organism of different species. This changes the genetic makeup of the organism that received the foreign gene.

✓ **Pathogenic organisms** are organisms that cause diseases.

∴ ***Answer D***

53. In which of the following ways is the carbon dioxide of the atmosphere fixed into the carbon found in organic molecules?
A .In the breathing processes of all animals.
B .In the decomposition of organic molecules.
C .In the process of photosynthesis by green plants.
D .In all the cellular respiration processes of organisms

Explanation :The **carbon dioxide** of the atmosphere is taken by the green plants through the opening known as **stomata** on the underside of the leaf. The carbon atoms in the carbon dioxide is reduced to the carbohydrate (organic molecules containing carbon) in the dark reaction of photosynthesis using energy and hydrogen atoms obtained in the light reaction of photosynthesis.

∴ ***Answer C***

54. Which one of the following characteristics is **UNNECESSARY** for an object to be considered a living thing?
A Ability to respond to stimuli
B .Ability to reproduce
C. Ability to grow.
D. Ability to move.

Explanation :The ability to respond to stimuli

- The ability to reproduce which enables living organisms perpetuate their species.
- Ability to grow from a single original cell-zygote in the case of sexual reproduction and from a group of actively dividing cells in a sexual reproduction (vegetative reproduction)are essential life processes that ensures the survival of the species.
- But albility to move is not the common characteristics of all organisms. Animals move from place to place for various life processes but plants do not.

∴ ***Answer D***

55. In which kingdom of life is it most likely to find many examples of organisms that can undergo metamorphosis?
A . Monera
B .Fungi
C. Plantae
D. Animalia.

Explanation: **Metamorphosis**, in zoology (animals study) marked changes of form during the life history of an animal occurs. Animals that grow in this way include many **fishes ,moluscs , insect , frogs etc.**

∴ ***Answer D***

56. Which term describes the process by which water is lost by evaporating through the stomata of leaves?
A .Transport
B .Transpiration
C. Anchorage
D. Adhesion

Explanation :

- **Transport** in living organisms refers to movement of materials from one part of the organism to the other part within a body.
- **Transpiration** is the loss of water vapor from the surface of a plant .Most water is the lost through stomata when they are open for gaseous exchange.
- **Anchorage:** The structure of an organism which help sit to fix (anchoring) or attach to something .For example, roots are anchoring structures that anchor or fix plants to the soil.
- **Adhesion:** Condition of touching but not growing together of parts normally separate; the specific binding together of cells, especially animal cells, by means of intercellular junctions and /or cell – surface molecules to form organized tissues

∴ ***Answer B***

57. What do you call structures that have the same evolutionary origin but now are different in structure or have different functions?
A .Endemic
B .Analogous
C. Homologous
D. Indigenous.

Explanation :

- **Endemic** :describing a population or species that is restricted geographically.
- **Analogous** are structural's of some organisms that function in very similar way. However morphologically and developmentally these structure are different .Because they are so different structurally, even though they have the same function, they cannot indicate the two species share a common ancestor.
- **Homologous** are structures that are very similar in form ,but very different in function. Because they are so similar they indicate an evolutionary relationships and a common ancestor of the species that possess them.
- **Indigenous** :describing an organism that is native to an area ,rather than introduced

∴ ***Answer C***

58. Which field studies the way in which individual traits of organisms are transmitted from one generation to the next?
A .Genetics
B .Ecology
C. Evolution.
D. Morphology.

Explanation :

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- **Genetics** :Scientific study of how physical, biochemical and behavior traits are transmitted from parents to their off spring.
- **Ecology (bio-ecology, environmental biology**):The study of the relationships of organisms to one another and to their surroundings.
- **Evolution:** in biology ,descent with modification, the process whereby all living things on earth have diverged by direct descent from a single origin of life that occurred more than 3,000 million years ago.
- **Morphology:** The study of the form of organisms

∴ ***Answer A***

59. What is a perfect flower?
A .A flower with petals and pistil.
B .A flower with stamens and sepals.
C .A flower with stamens and pistil.
D .A flower with sepals and petals.
Explanation :The majority of flowers possess both **stamens** and **carpel's (pistils)**.Such a flowers are said to be **perfect flowers.**

∴ ***Answer C***

60. Which region of the human digestive tract contains enzymes that perform well at low PH?
A .Mouth C. Small intestine
B .Stomach D. Large intestinal
Explanation:

- **Mouth :** The PH in the mouth varies from being slightly alkaline (PH 7.5) to quite acidic (PH 5.0) depending on whether or not we have eaten and so what we have eaten.
- **Stomach:** The PH in the stomach can be as low as PH 1.5.
- **Small intestine**: The PH of the small intestine is slightly alkaline at PH 7.5.
- **Large intestine:** Digestive enzymes from different regions have optimum PHs that reflect the region in which they are secreted. There are no secretions in the large intestine. The PH of this tract depends on the PH of substances that pass down into it from the small intestine.

∴ ***Answer B***

61. Which of the following resources is needed by an ecosystem from outside in order to sustain itself?
A .Producers C. Micronutrients.
B .Macronutrients D. Sunlight.
Explanation :

- **Producers** of the ecosystem are the bases of the ecosystem that produces organic food from inorganic substances using light energy or chemical energy (as in chemotrophs). They are the primary source of food for all biotic component of the ecosystem.

- **Macro nutrient** are nutrients that are required in larger amounts that are obtained as a result of decomposition by micro-organisms. They circulate in the ecosystem.
- **Micro nutrients** are nutrients that are needed in smaller amounts and obtained as a result of decomposition by micro-organism. They circulate in the ecosystem.
- **Sun light** is the ultimate source of energy for all living systems in the ecosystem. It is the only resources obtained from outside the ecosystem.

∴ ***Answer D***

62. Which of the following is prevented from taking place if populations are separated by a geographic barrier?
A .Mutation C. Gene flow
B .Evolution D. Natural selection
Explanation :As long as two populations are able to inter breed (genes of the two populations flow),they are unlikely to evolve into distinct species. They must somehow go through a period when they are prevented from inter breeding (prevent gene flow between them).Both allopathric speciation (speciation by isolation or separation by geographical barrier)and sympatric speciation (speciation that need not involve physical separation)involve isolation mechanisms that prevent different population from inter breeding (prevent gene flow) for a period of time.

∴ ***Answer C***

63. For a biological detergent to effectively remove oily and greasy dirt, which of the following enzymes should it contain?
A . Cellulase B . Protease C. Amylase D. Lipase
Explanation:

- **Cellulase :** an enzyme that breaks cellulose molecules in to glucose like molecules.
- **Protease** used for pre-Soak and main wash to remove protein residues.
- **Amylase** used for pre-soak and main wash to remove starch residues.
- **Lipase** is used to ripening of Danish blue cheese and in detergents. It is used to remove oil and greasy dirt's in main washing and pre-soak washing.

∴ ***Answer D***

64. If a cell fails to clear its cellular debris, Which one of its organelles is most likely **NOT** functioning?
A .Nucleus
B .Mitochondria C. Endoplasmic reticulum
Explanation: D. Lysosomes.

- **Nucleus :** a large dense organelle bounded by a double membrane (nuclear envelop)present in eukaryotic but not prokaryotic cells, and

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which contained the chromatin and in which DNA replication and transcription take place. It controls all the activities of the cell.

- **Mitochondria** are organelles in the cytoplasm of eukaryotic cells ,having a double membrane. They are the sites of aerobic respiration.
- **Endoplasmic reticulum(ER):** Extensive convoluted internal membrane in eukaryotic cells, continuous with outer nuclear membrane and enclosing a continuous internal space (lumen).Involved in the synthesis and transport of membrane proteins and lipids and of proteins destined for secretion from the cell.
- **Lysosomes** are organelles surrounded by a single membrane. They are formed in the Golgi apparatus and contain digestive enzymes that break down cellular wastes and debris.

∴ ***Answer D***

65.Which of the following applies to the tropical rainforest?

A .Low rainfall and high temperature.
B .Low rainfall and low temperature.
C .High rainfall and high temperature.
D .High rainfall and low temperature.

Explanation: **Tropical rain forest** is one of the most complex and species – rich ecosystem in the world. Since it occurs near the equator, the warmth and high moisture due to abundant rain fall allow a continuous growth of plants.

∴***Answer C***

66. Which of the following is true about saturated fatty acids?

A .They have single bonds between carbon atoms.
B .They are the same as polyunsaturated fatty acids.
C .They are the same as monounsaturated fatty acids.
D .They have double bonds between the carbon atoms.

Explanation: **Fatty acids** and lipids lacking double bonds (C = C) are said to be saturated fatty acid.

∴ ***Answer A***

67.During protein synthesis, where in the cell does transcription take place?

A. Ribosome
B .Nucleus
C. Endoplasmic reticulum.
D. Golgi apparatus.

Explanation: **In eukaryotic cells** :Transcription and translation are separated:

- transcription occurs in the **nucleus.**
- translation occurs in the **cytoplasm.**
- eukaryotic mRNA are modified before leaving the nucleus.

∴ ***Answer B***

68.Which molecules carry the instructions for protein synthesis?

A .Carbohydrates & lipids
B .Amino acids
C.DNA &RNA
D. Enzymes

Explanation: **DNA (deoxyribo nucleic acid)** is the molecule that stores the genetic information. This genetic information which is the code for protein synthesis is specified by DNA and has to be carried to the ribosomes so that they can assemble the amino acids in the correct sequence to form the proteins. These codes instruction for synthesis of a protein are carried by molecules of RNA (Ribo nucleic acid) Particularly messengers RNA (m RNA).

∴ ***Answer C***

69.In protein synthesis, what is produced during transcription?

A .Protein
B .mRNA
C.DNA
D. Polypeptide.

Explanation: **Protein synthesis** involves two main stages. The first stage of protein synthesis is called **transcription.** In this process the DNA code for protein synthesis is rewritten in molecule of messenger RNA (m RNA) .This rewriting of the code is called transcription.

✓ In the seconds stage of protein synthesis the mRNA moves out of the nucleus into the cytoplasm and attaches to the ribosome (site of protein synthesis).The code message from the DNA that were rewritten in the mRNA are interpreted to specific amino acids that join to form a polypeptide chain or protein.

∴ ***Answer B***

70. Which of the following groups of plants carry out light dependent and light independent reactions of photosynthesis in separate cells?

A .C- 3 plants
B .C -4 plants
C.CAM plants
D. Plants without chlorophyll

Explanation:

✓ **The three – carbon cycle** or the calvin cycle (C- 3).
✓ The reduction of carbon takes place in the stroma in a cycle called the calvin cycle named after its discovered ,Melvin calvin .The starting and ending compound is a five carbon sugar ribulosediphosphate (RuDP).
✓ The process begins when carbon dioxide enters the cycle and is bound to RuDP ,which then splits to form two molecules of phospho glycerate (PGA) .Each PGA molecule contains three carbon atoms. Hence the name ,the three – carbon path way. At each full turn of the cycle, a molecule of carbon dioxide enters the cycle ,is reduced and a molecule of RuDP is generated. Six revolution of the cycle, with the introduction of six atoms of carbon, are necessary to produce a six carbon sugar, such as glucose.

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❖ **C- 4 Plants** or the four carbon photosynthetic path way. In some plants the first product of carbon dioxide fixation is not three- carbon molecule phosphoglycerate, but the four-carbon compound oxaloacetic acid. Plants that utilize this path are commonly called the C-4 ,or four – carbon plants.

✓ In this plants the light dependent reaction occurs in mesophyll cells and the light independent reaction occurs in the bundle-sheath cells.

❖ **CAM-plants :**cacti are desert plants that carry CAM photosynthesis. Cacti uses what is essentially the same set of reactions as C_4 plants, but they separate the two stages not by carrying them out in different cells, but by carrying them out at different times (night and day).

∴ ***Answer B***

71. Which stage in the life cycle of HIV is inhibited by the antiretroviral drug known as integrase – inhibitor?
A .Conversion of viral RNA to DNA.
B .Replication of the viral genome.
C .Insertion of the viral genome into host genome.
D .Release of the viral progeny from the cell.
Explanation: In the life cycle of the HIV - particle ,after the virus entered the host cell, the enzyme reverse transcriptase converse its RNA in to DNA.

✓ This viral DNA becomes integrated in to the cell own DNA where it is transcribed to viral RNA that starts production of viral proteins. Therefore, any antiretroviral drug that inhibits the action of enzyme which integrate viral DNA in to the host cell DNA (integrate inhibitor)inhibits the insertion of viral genome in to host genome.

∴ ***Answer C***

72. Which of the following pairs of individuals have identical DNA finger prints?
A .Father and son. C. Brother and sister
B .Mother and daughter D. Monozygotic twins.
Explanation: Different individual members of any family that are born at different time or the same time(in the case of fraternal twins) are produced from different egg cells and sperm cells which are genetically different. Therefore, they always have different genetic **finger prints.**

✓ The only two individual that can have identical DNA finger prints are monozygotic twins or identical twins ,because these are produced from the same egg cell and sperm cell.

∴ ***Answer D***

73. The half life of carbon – 14 is about 5730 years. If a fossil is 17200 years old, about what percent of its original carbon- 14 is still present in the fossil?
A .75% B.50% C .25% D.12%
Explanation: At the end of the first half life (5730 years) only 50% othe origin carbon- 14 remains.
• At the end of the second half – life (11460 years) only 50% the remaining or 25% of the original carbon -14 is left.
• At the end of the third half – life (17200 years) only 50% of the remaining or 12.5% of the original carbon – 14 remains.

∴ ***Answer D***

74. Among the following identify the organelle in which nucleic acid is NOT found?
A .Mitochondria C. Ribosome
B .Chloroplast D. Golgi apparatus.
Explanation:
❖ **Mitochondria** contain their own DNA and different types of RNA (nucleic acids).
❖ Similarly **chloroplasts** also have their own DNA and RNA (nucleic acids).
❖ **Ribosome's** are made from ribosomal RNA (nucleic acid) and protein units
❖ **Golgi apparatuses** do not contain any nucleic acid.

∴ ***Answer D***

75. Suppose sites A,B,C and D have Simpson's index of diversity values of 20, 8, 16 and 3 ,respectively, which site is dominated by one or just a few species?
A .Site A B .Site B C. Site C D. Site D
Explanation: **Site A** which has simpson's index of diversity value of 20 as compared to others sites (B ,C and D) is considered as an area with a highest diversity index. A higher diversity index suggests a number of successful species and a more stable ecosystem.

✓ **Site D** with diversity index value of 3 is a site having a lowest ,value for the index of diversity.

✓ A low value for the index of diversity suggests an area dominated by one or just a few species.

∴ ***Answer D***

76. Suppose a hypothetical cube-shaped cell has sides of 10 micrometers, what is the surface area to volume ratio of this cell?
A .6 :10 C.3 :6
B .10 :10 D .4 : 8

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Explanation: The area of one side of the cell is 10 micro meter × 10 micro meter = 100 micro meter^2.

The surface area of the cell is $100\mu m^2 \times 6 = 600\mu m^2$

The volume of the cell is $10\mu m \times 10\mu m \times 10\mu m = 1000\ \mu m^3$

The surface are to volume ratio of the cell is $\frac{600}{10,00} = 6:10$

∴ ***Answer A***

77. Which of the following can create strong inter-specific competition if shared by two or more species?

A .Habitat
B .Ecosystem
C. Niche
D. Predator.

Explanation: **A habitat** is an area where a population lives and finds the nutrients , water living space and other essential resources it needs to survive.

Sometimes ,different organisms can share the same habitat. However, they make different demands on that habitat. This combination of habitat and the demands made is called the **ecological niche.** The ecological niche of an organism describes its role within a habit.

If two or more species occupy the same niche , they make the same demand, this create a strong inter-specific competition (competition between different species)where one of the two species out competes the other, which may die out.

∴ ***Answer C***

78.Among the following ,which one has little or no contribution to the loss of biodiversity?

A .Overgrazing by domestic animals.
B .Clearing forests to grow crops.
C .Growing crops of improved varieties
D .Growing crops of landraces.

Explanation: The following are practices with direct effect on loss of biodiversity!

- **Overgrazing by domestic animals :** reduces the availability of forage (animal food)and woody plant species for the other animals.
- **Clearing of forests to grow crops (deforestation):** conversion of forest ,wood lands and savannas to agricultural lands and other land use system reduces the area available to native species. Destruction of the native species reduces the biodiversity of the area.
- **Growing crops of improved varieties** introduction of improved crop verities – reduces the genetic diversity of the particular plants as only the improved variety is used.
- **Growing crops of land races :** means growing crops or plants indigenous to the area maintains the biodiversity of the crops.

∴ ***Answer D***

79. Which of the following types of chromosome mutations gives rise to an extra chromosome?

A Inversion
B .Translocation
C. Deletion
D. Non-disjunction.

Explanation:

- ***Inversion :*** This is a type of chromosomal mutation which occurs when an area of DNA on the chromosomes reverse its orientation on the chromosome. Just one inversion on the chromosome 16 can cause leukemia.
- **Translocation** is a type of chromosomal mutation where a piece of one chromosome is transferred to another non-homologous chromosome .This type of chromosome mutation is often responsible for chronic myelogenous leukemia.
- **Deletion** is another type of chromosomal mutation where a decrease in the number of genes occur due to the deletion of a large section of a chromosome. Deletion can result in a variety of genetic disorders, such as prader – Willi syndrome. This results from a malfunctioning of the hypothalamus.
- **Non- disjunction**: when homologous chromosomes do not separate successfully to opposite poles during meiosis , the result is one of the gamete lacking chromosomes (receiving less chromosome number) and the other having an extra chromosome.

∴ ***Answer D***

80.Among the following fossil hominid species, which one is the oldest of all?

A .homo erectus.
B . Ardipithecus ramidus
C. Australopithecus afarensis
D. Australopithecus africanus.

Explanation:

- The fossil of Homo erectus is 1 to 2 million years old.
- Fossil of Ardipithecus ramidus is 4.4 million years old.
- Fossil of Australopithecus afarensiss (Lucy) is 3.2 million years old.
- Fossil of Australopithecus Africanus is 2 to 3 million years old.

∴ ***Answer B***

81.During aerobic respiration ,what is the route through which protons return from the mitochondrial inter- membrane space back to its matrix?

A .Proton pump
B .ATP synthase
C. Ion channel
D. Membrane lipid.

Explanation: In aerobic respiration the electrons released from the oxidized NAD and FAD pass along a series of electron carriers that form the electron transport chain. The electrons lose energy as they pass from one carrier to the next. This powers the proton pumps that move

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protons from the matrix of the mitochondrion to the inter-membrane space. Because of the action of the proton pumps, protons accumulate in the inter membrane space creating a higher concentration than in the matrix. This proton gradient results in proton diffusion from the inter membrane space back to the matrix through the ATP synthase molecule resulting the synthesis of ATP from ADP and P1.

∴ ***Answer B***

82. Which of the following mating produces children al having the same ABO blood phenotypes?

A .AO× AB
B .BB× OO
C. AO ×BO
D.AA ×BO

Explanation:

❖ **AO ×AB**

- Parental phenotype, A blood type and Ab blood type .
- parental genotype : $I^A I^O \times I^A I^B$.
- Gametes : $I^A, I^O \times I^A, I^B$.
- Fertilization:

	I^A	I^O
I^A	$I^A I^A$	$I^A I^O$
I^B	$I^A I^B$	$I^B I^O$

- $\frac{1}{2}$ (50%) of the children will have A blood group.
- $\frac{1}{4}$ (25%) of the children will have AB blood type.
- $\frac{1}{4}$ of (25%) of the other children will have B blood type.

❖ **BB× OO**

- Parental blood type B and O.
- Parental genotype: $I^B I^B \times I^O I^O$.
- Gametes..... I^B and I^O
- Fertilization $I^B \times I^O \Rightarrow$ off spring $= I^B I^O$
- All of the children will have the same blood type B.

❖ **AO × BO**

- parental phenotype (blood type) :A and B.
- parental genotype : $I^A I^O \times I^B I^O$
- Gametes : I^A, I^O and I^B, I^O.
- Fertilization:

	I^A	I^O
I^B	$I^A I^B$	$I^B I^O$
I^O	$I^A I^O$	$I^O I^O$

- All of the four blood type A,B ,AB and O will be possessed by the children.

❖ AA × BO

- Parental blood types (phenotypes) : A and B.
- Parental genotypes : $I^A I^A$ and $I^B I^O$.
- Gametes ..... I^A and I^B, I^O.
- Fertilization

	I^B	I^O
I^A	$I^A I^B$	$I^A I^O$

- $\frac{1}{2}$ (50%) of the children will have AB blood type.
- $\frac{1}{2}$ (50%) of the other children will have A blood type.

∴ ***Answer C***

83. On which of the following does the rate of algal photosynthesis in a lake depend?

A .The oxygen content of the water.
B .The nitrogen content of the water.
C .The elevation where the lake is found.
D .The amount of light that penetrates the lake water.

Explanation: The rate of photosynthesis of the algal population in a lake depends on the amount of light penetrating the water of the lake.

∴ ***Answer D***

84. Choose the one that represents a chemical formula of an organic compound?

A .$C_{12}H_{22}O_{11}$
B. $Mg(OH)_2$
C . NH_3
D. NaCl

Explanation:

- $C_{12}H_{22}O_{11}$ is an organic compound because it contains both carbon and hydrogen atoms.
- $Mg(OH)_2$ is an inorganic compound (is not organic) because it does not contain carbon atoms.
- NH_3 is an inorganic compound (is not organic) because it does not contain carbon atom.
- NaCl is an inorganic compound (is not organic) because it does not contain carbon and hydrogen atoms

∴ ***Answer A***

85. Which of the following is the largest ecological unit?

A .A community
B .A biome
C. A population.
D. An ecosystem

Explanation:

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- **A community** is composed of all the populations in a given area. In other words a community refers to the living component of the ecosystem.
- **A biome** is geographical or regional area with specific climate ,a specific soil type and specific animals and plants that are adapted in similar ways to abiotic conditions within the area.
- **A population** is a group of closely related organisms that normally interbreed (the same species) and live together.
- **An ecosystem** is a natural unit in which energy flows from the source to producers, consumers and decomposers and nutrients circulate from the soil through plants, animals and back to the soil through the action of micro organisms.

∴ ***Answer D***

86. Which of the following pairs are analogous structures?
A .Wing of a bird & wing of a butterfly.
B .Front leg of a horse & a human arm.
C .Wing of a bird & wing of a bat.
D .Front leg of a frog & Wing of a bat.
Explanation: Analogous structures are structures that are different morphologically ,developmentally and evolutionary origin but having the same function.

∴ ***Answer A***

87. From which of the processes of cellular respiration is the majority of the ATP generated?
A .Anaerobic fermentation
B . Glycolysis and link reaction.
C .Electron transport and chemiosmosis.
D .Krebs cycle and glycolysis
Explanation:
- **Anaerobic respiration** generates only 2 ATP molecules.
- **Glycolysis** is the anaerobic stage of aerobic respiration.It generates only 2ATP molecules. Link reaction does not directly generates ATP but it produces NAD that later on generates ATP.
- **Electron transport and chemiosmosis** generate 34 ATP molecules.
- **Krebs cycle generate** 2 ATP and **glycolysis** 2 ATP.

∴ ***Answer C***

88. In which of the following solutions does an animal cell undergo haemolysis?
A .In hypotonic solution.
B .In hypertonic solution.
C .In isotonic solution.
D .In both hypotonic & hypertonic solutions.

Explanation: **Hypotonic solution:** If a cell is surrounded by pure water , or a solution whose solute concentration is lower than that of the cell's content ,water molecules pass into the cell by osmosis and such a solution is said to be **hypotonic to the cell.** The cell swells and becomes turgid, if it is a plant cells. Animals cells kept in hypotonic solution swells and burst or lysed (because of absence of cell wall).This condition is called **haemolysis** and the cell is said to be **haemolysed**.

∴ ***Answer A***

89. How are unsaturated fats distinguished?
A .They are made up of glucose & fructose.
B .They are made up of amino acids & glycerol.
C .They have double bonds in their carbon chains.
D .They have only single bonds at the points of attachment.
Explanation: **Unsaturated fats** are fats composed of fatty acids that have double bonds in their carbon chains.

∴ ***Answer C***

90. What is the difference between HIV and AIDS?
A .AIDS is the virus and HIV is the disease.
B .Both AIDS and HIV refer to the disease.
C .AIDS is the disease and HIV is the virus.
D .Both AIDS and HIV refer to the virus.
Explanation: **HIV (Human Immuno deficiency virus)** a virus that infect and destroy the immune system of the body and finally leading to the diseases known as AIDS (Acquired immuno deficiency syndrome).

∴ ***Answer C***

91. What is the ecological succession in which plant communities establish after an area has been completely demolished by fire?
A .Tree succession
B .Secondary succession
C. Primary succession.
D. Bare rock succession.
Explanation: **Secondary succession**: A plant succession following the interruption or destruction of the normal climax community by fine or other means of deforestation.
- ✓ **Tree succession and secondary succession** are the same events following destruction of a forests.
- ✓ **Primary succession:** a plants succession that begins on a bare ground (this could be a bare rock)

∴ ***Answer B***

92. Which of the following conclusions can be drawn from the structural similarities observed between the flippers of whales and the arms of humans?
A .The human species began life in the ocean.
B .The human species and whales share a common ancestry.

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C .Whales are older than the human species.
D .Whales have evolved from early humans that went back to the ocean.
Explanation: Some organisms have anatomical structures that are very similar in form, but very different in function. We call such structures homologous structures. Because they are so similar, they indicate an evolutionary relationships and a common ancestor of the species that possess them. The flippers of whales and the arms of humans are homologous structure that show human and whales share common ancestor.

∴ ***Answer B***

93. Which of the following is produced when a lipid molecule is hydrolyzed?
A .Amino acids & water.
B .Amino acids & glucose
C. Fatty acids & glycerol
D. Glucose & glycerol.
Explanation: When lipids molecules are hydrolyzed by hydrolytic enzyme they produce fatty acid and glycerol molecules.
- The hydrolysis of **protein** produces **amino acids** .
- The hydrolysis of **carbohydrate** produce **glucose**.

∴ ***Answer C***

94. Which of the following did the early heterotrophs do that probably assisted the origin of autotrophs?
A .They added oxygen and chlorophyll to the environment.
B .They added carbon dioxide to the environment.
C .They stored energy in the bonds of inorganic compounds.
D .They manufactured food from carbon dioxide and oxygen.
Explanation: According to the endosymbiont theory that suggests the origin of eukaryotes ; the prokaryotic membranous cell that engulfed smaller cell that respire aerobically engulfed, other smaller prokaryotic cell. These were prokaryotic cells that could photosynthesis and would, in time evolve in to chloroplast. The cells that contained these would be autotrophs and the fore runners of plant cells. The earliest prokaryotes that engulfed other smaller cells that were better at respiring organic molecules became heterotrophs with efficient aerobic respiration that released more energy and carbon dioxide from organic compounds.

∴ ***Answer B***

95 .Suppose in an experiment conducted to test a new drug, 20 people were given the drug and 20 People a placebo. What is the control of the experiment?
A .People who received the placebo.
B .People who received the new drug.
C .The new drug itself.
D .The new drug and the placebo.

Explanation: A **control group** acts as a **standard** for comparison .It is used to **isolate** the factor we are investigating and show that changes are due to this factor. For example ,in drug trials ,one group of people with the condition the drug is used to treat is given a tablet containing the drug (the experimental group). Another group is given a placebo – a tablet containing no drug (the control group).If both groups get better , then its seems that the drug is having little effect. If only the experimental group get better , this must be due to the drug .Without the control group we would not have been sure.

∴ ***Answer A***

96. If a DNA contains 10% thymine,what is the percentage of cytosine in the DNA?
A .10%
B.30%
C.40 %
D.90%
Explanation: According to the base – pairing rule in any DNA molecule, **thymine** pairs with **adenine** and **cytosine** pairs with **guanine**. Therefore , the number or percentage of thymine is equal to that of adenine and the number or percentage of cytosine is equal to that of guanine. In a DNA molecule that contains 10% thymine , there is also 10% adenine .Out of the remaining 80% of bases , 40 % is cytosine and 40 % is guanine .

∴ ***Answer C***

97. Which of the following will happen if a plant cell is kept in a solution that is stronger than its protoplasm?
A .The cell will become turgid.
B .The central vacuole will expand
C .The cell protoplasm will get plasmolysed.
D .The cell will swell and burst.
Explanation: A solution stronger than the protoplasm of the cell always contains more solute particles and less water molecules than the protoplasm of the cell. Since water is moving from its higher concentration area to lower concentration area ;water moves out of the cell in to the stronger solution. Therefore, the protoplasm (the cell)loses its turgidity which results a plasmolysed cell (protoplasm).

∴ ***Answer C***

98. What does the absence of chloroplasts in a unicellular organism indicate?
A .Reproduction by sexual means.
B .Autotrophic nutrition.
C .Heterotrophic nutrition.
D .Complete lack of metabolic processes.
Explanation: If **Chloroplast** is absent in a unicellular organism, the cell cannot manufacture its own organic food from inorganic carbon dioxide

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and water. This means that it cannot photosynthesis .Such cells are said to be heterotrophic.

∴ ***Answer C***

99. From which organism is the first antibiotic used in medicine produced?
A .Gram negative bacteria.
B .Members of the fungi.
C .Genetically modified protozoa.
D .Higher plants.

Explanation: It was in 1928, a young researcher known as **Alexander Fleming** in UK found the first antibiotic, which he called it penicillin was produced by a common mould called **penicillium notatum.**

∴ ***Answer B***

100. Biotechnology can be applied in environmental remediation. To which of the major application areas of biotechnology is this most related?
A .Medicine and nutrition.
B .Medicine and industry.
C .Biodiversity and industry.
D .Agriculture and environment.

Explanation: **Genetic engineering** could increase genetic diversity (Biodiversity), and produce more variant alleles which could also be crossed over and implanted in to other species.

✓ **Genetic engineering (the modern biotechnology)**is much quicker process in creating genetic diversity (biodiversity) than traditional breeding.

∴ ***Answer C***

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